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DE L'ESPACE

Advanced Master - Space Systems
Engineering (TAS-ASTRO)
End of Study Internship - Report

OPTICAL CHARACTERIZATION FOR OPTIMIZING
MICROWAVE KINETIC INDUCTANCE DETECTOR (MKID)
TECHNOLOGY FOR USE IN ASTRONOMICAL OBSERVATIONS

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Introduction

This report documents the six-month internship I completed as part of my Advanced Master in Space Systems Engineering TAS-ASTRO at ISAE-SUPAERO. My internship took place at the California Institute of Technology (Caltech) in the Physics, Mathematics, and Astronomy Division, specifically within the Astronomy and Observational Cosmology research department.

During this period, I had the chance to work with a highly skilled and dedicated team focused on cutting-edge research in observational cosmology. Under the mentorship of Professor Sunil Golwala, I was involved in developing a prototype of a phonon-mediated superconducting detector aimed at the direct detection of dark matter and in the development of the Next Extended Wavelength Multi-band Sub-millimeter Inductance Camera (NEW MUSIC). My specific internship objective (and the focus of this report) was the optical characterization and improvement of Microwave Kinetic Inductance Detectors (MKIDs), which are integral to the performance and sensitivity of the NEW MUSIC instrument. This innovative project has the potential to significantly enhance our understanding of the universe. More precisely the scientific motivation behind this specific project is in the maximization of the resources. As telescope's aperture size tends to increase, the focal plane real estate becomes more expensive, implying a strong need in optimizing the use of the focal plane area in millimeter and sub-millimeter bands.

Pr. Golwala is a distinguished professor of physics at Caltech and serves as the Deputy Executive Officer for Astrophysics. He earned his B.A. from the University of Chicago in 1993, followed by his M.A. and Ph.D. from UC Berkeley in 1995 and 2000, respectively. Pr. Golwala's research is primarily centered on the development of advanced detectors for astrophysical applications. He has made significant contributions to the field, particularly in the areas of dark matter detection and sub-millimeter astronomy.

Pr. Golwala has been involved in several groundbreaking projects, including the development of Weakly-Interacting Massive Particle (WIMP) dark matter detectors using Microwave Kinetic Inductance Detectors (MKIDs). He has also worked on the Multi-band Submillimeter Inductance Camera (MUSIC), a sophisticated instrument (which NEW MUSIC will be the successor) designed to succeed Bolocam on the Caltech Sub-millimeter Observatory (CSO) and serve as a technology demonstrator for the Cornell-Caltech Atacama Telescope (CCAT). His work on the Cosmic Microwave Background (CMB) polarization experiments, such as SPIDER and BICEP2/Keck Array, highlights his extensive expertise in observational cosmology.

The team I worked with at Caltech is renowned for their expertise and groundbreaking research.

- Dr. Fabien Defrance: A Research Scientist at NASA's Jet Propulsion Laboratory, bringing expertise and insights into the projects I worked on.
- Dr. Jean-Marc Martin: A PostDoctoral Fellow Researcher who contributed significantly to my project.

- Shilling Yu: A visiting graduate student from the University of the Chinese Academy of Sciences in Shanghai, currently pursuing a PhD in Physics.
- Sam Condon: A former Post-Baccalaureate on NEW MUSIC who is now a graduate student at Stanford University.
- Yann Sadou (myself): Employed as a visiting student and Post-Baccalaureate Researcher.

Their collective knowledge and passion for observational cosmology made this internship an incredibly enriching experience.

Throughout my time at Caltech, I engaged in various tasks, from theoretical modeling and simulations to hands-on experimentation and data analysis. This multifaceted approach allowed me to gain a comprehensive understanding of the project's scope and the challenges involved in developing such advanced detection technologies.

More concretely each week the team had one meeting to talk about advances in each parts of our projects. These were good opportunities for me to share my updates but also my encountered problems and receive advices as well as help in calculation and decisions to make. The rest of the time I was mainly staying in full autonomy whether it be during the simulation, the design or the construction. In terms of resources, I had access to a computer and all the laboratory capabilities especially during the phases of construction and measurements. Financially, the money involved in this project came from grants from NSF and NASA allocated by the principal investigator to NEW-MUSIC.

This report will outline the objectives, methodologies, results, and personal reflections from my internship experience. I am immensely grateful for the opportunity to work with such a talented and inspiring group of individuals, and for the support and guidance they provided throughout my stay at Caltech. This experience has been invaluable in shaping my skills and knowledge, and it has solidified my passion for a career in space systems engineering.



Figure 0.1: Caltech Sub-millimeter Observatory (CSO) at night

1 Introduction to the Microwave Kinetic Inductance Detector (MKID)

Microwave Kinetic Inductance Detectors (MKIDs) are cutting-edge devices that represent a significant advancement in the field of observational cosmology and astrophysics. These detectors are particularly crucial for observing faint and distant astronomical objects across a broad spectrum of electromagnetic radiation. As the sensitivity and resolution of astronomical observations continue to push the boundaries of what is observable, MKIDs have emerged as a critical technology for next-generation instruments. This section provides a detailed overview of the science behind MKIDs, their history, operational principles, and their applications in astrophysics.

1.1 Historical Development of Kinetic Inductance Detectors (KIDs)

The concept of kinetic inductance, the underlying principle of MKIDs, was first described in the mid-20th century as researchers explored the properties of superconducting materials. The journey from theoretical understanding to practical application in astronomy, however, took several decades. Here are the main important steps in the development of this technology:

Early Theoretical Foundations:

- **The Discovery of Superconductivity (1911):** The phenomenon of superconductivity was first discovered by Heike Kamerlingh Onnes in 1911, when he observed that mercury exhibited zero electrical resistance at very low temperatures. This discovery laid the groundwork for the exploration of superconducting materials and their unique properties.[1]
- **Kinetic Inductance Concept (1960s):** The concept of kinetic inductance emerged in the 1960s, when researchers began to understand that the inertia of Cooper pairs in a superconductor could contribute to the material's inductance. This understanding was primarily theoretical at first, with limited practical applications. [2]

Development of Kinetic Inductance Detectors:

- **First Proposals (1980s-1990s):** In the late 20th century, scientists began exploring the idea of using kinetic inductance as a basis for detectors. These early proposals suggested that changes in kinetic inductance could be used to detect photons, particularly in the millimeter and sub-millimeter wavelengths.
- **Initial Experiments (Early 2000s):** The first practical demonstrations of KIDs were conducted in the early 2000s. These experiments confirmed that kinetic inductance could be harnessed to create highly sensitive detectors for astronomical observations.

Transition to MKIDs:

- Introduction of MKIDs (2003): In 2003, Peter K. Day and his colleagues at Caltech introduced the concept of Microwave Kinetic Inductance Detectors (MKIDs). This innovation involved using superconducting resonators to detect changes in kinetic inductance caused by photon absorption. MKIDs represented a significant advancement over previous detector technologies, offering higher sensitivity, lower noise, and the ability to be easily multiplexed.[3]
- First Astronomical Applications (Mid-2000s): Following their introduction, MKIDs were quickly adopted in various astronomical instruments such as NIKA 1, NIKA 2 or SPIAKID where France was involved. The first successful astronomical observations using MKIDs were conducted in the mid-2000s, marking the beginning of their widespread use in astrophysics as we can see on figure 1.1. [4]

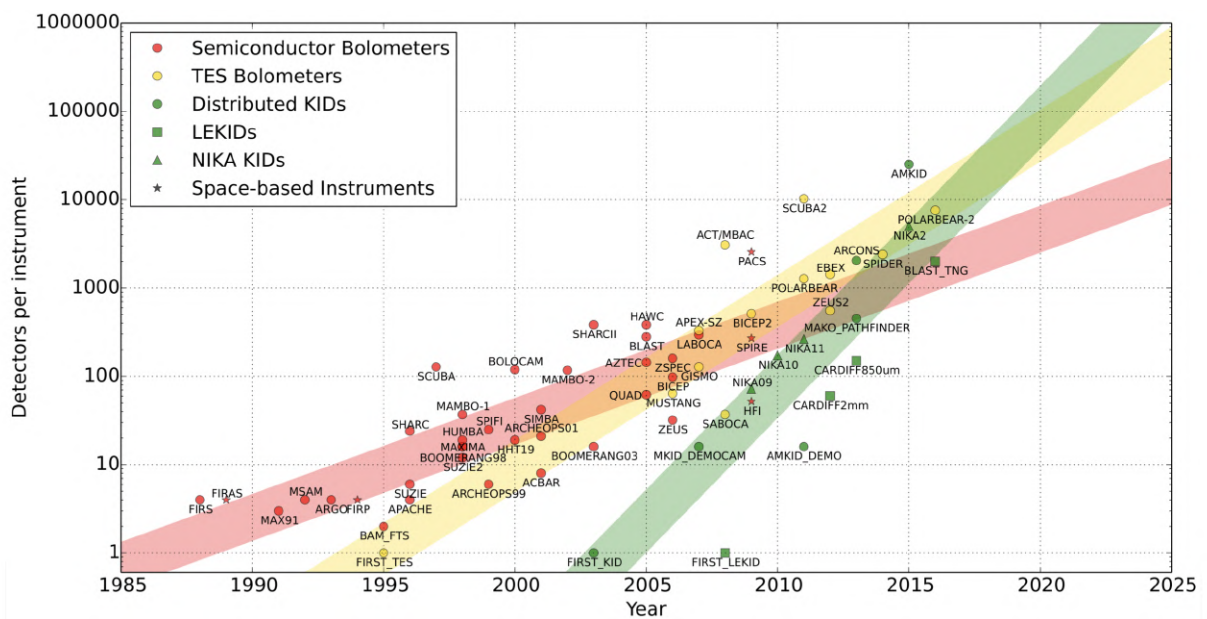


Figure 1.1: Evolution of astronomy detectors over years [4]

1.2 Principles of MKIDs

The KIDs rely on the unique properties of superconductors, which allow for the highly sensitive detection of electromagnetic radiation. The principles behind KIDs involve sophisticated concepts in superconductivity, resonant circuits, and frequency multiplexing, making them ideal for large-scale detector arrays.

The Superconducting State and Kinetic Inductance:

Superconductors exhibit zero electrical resistance when cooled below a critical temperature (T_c). This behavior arises because electrons in a superconductor pair up to form Cooper pairs, which move through the lattice without scattering, thereby eliminating resistance.[5] However, while superconductors have zero DC resistance, they do exhibit a non-zero impedance to alternating current (AC), primarily due to kinetic inductance. Kinetic Inductance occurs because Cooper pairs have mass and inertia, which means they

take time to accelerate when an electric field is applied. This leads to the storage of kinetic energy in the moving Cooper pairs, which contributes to the overall inductance of the material. Additionally, a magnetic field penetrates the superconductor only up to the London penetration depth (λ), which also contributes to the surface inductance. The surface impedance Z_s of a superconductor is expressed as:

$$Z_s = R_s + i\omega L_s$$

where R_s is the surface resistance, ω is the angular frequency, and L_s is the surface inductance. At temperatures much lower than T_c , R_s is negligible compared to ωL_s , meaning the impedance is predominantly inductive. [3]

Interaction with Photons: Breaking Cooper Pairs

When a photon with energy greater than twice the superconducting gap energy (2Δ) is absorbed by the superconductor, it breaks a Cooper pair, generating quasi-particles (unpaired electrons). The number of quasi-particles created by a photon is approximately:

$$N_{qp} \approx \eta \frac{h\nu}{\Delta}$$

where $h\nu$ is the energy of the photon and η is the efficiency with which the photon's energy is converted into quasi-particles and which depends on the material. These quasi-particles increase the surface impedance of the superconductor by affecting both the inductive (L_s) and resistive (R_s) components. The kinetic inductance L_s increases because the density of Cooper pairs decreases, while L_s increases due to the presence of quasi-particles. The following figure 1.2 illustrates the principle of detection.

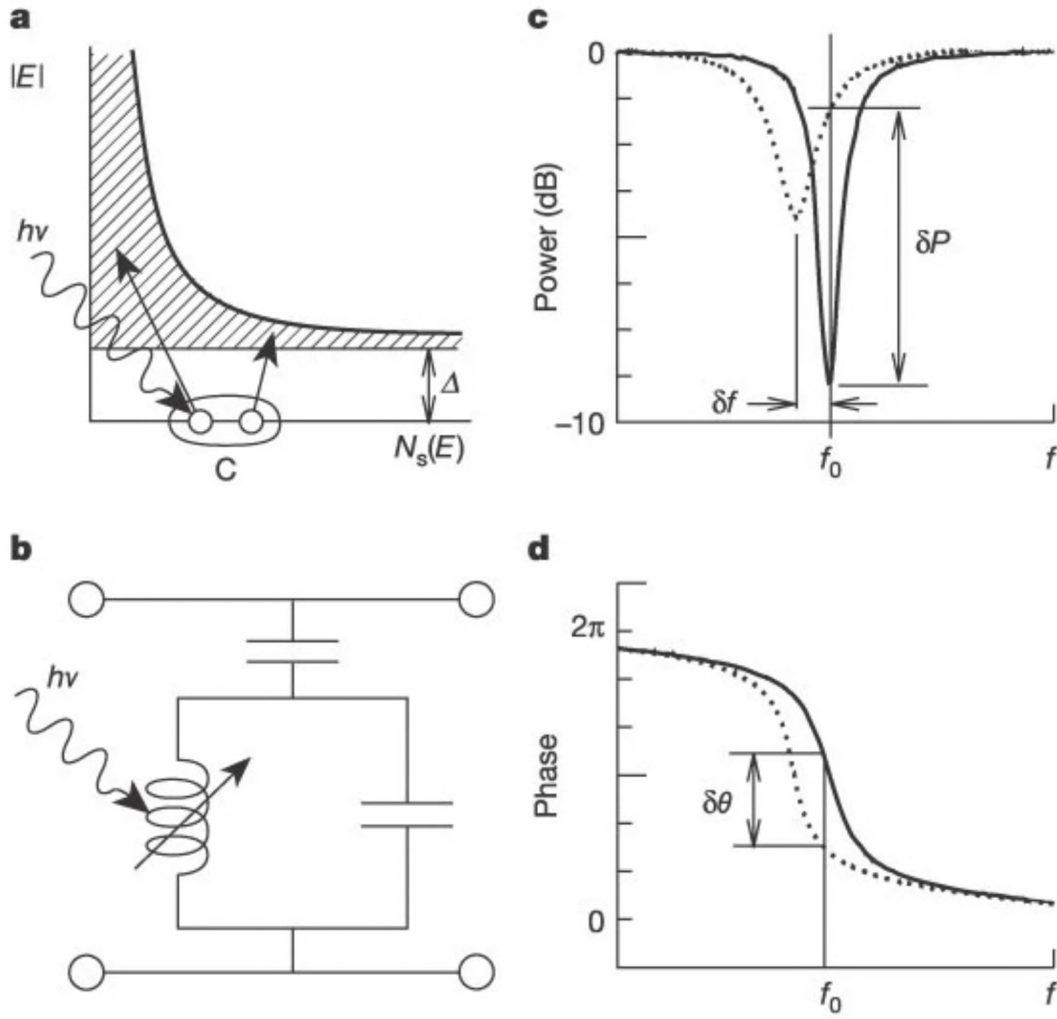


Figure 1.2: Illustration of the detection principle [3]

KIDs are typically designed as superconducting resonators, where the inductive component of the circuit L is the sum of the kinetic inductance L_s and the geometric inductance L_g (directly influenced by the design of the superconductor): $L = L_s + L_g$. The resonator's frequency is determined by:

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

Where L is the total inductance and C is the capacitance. We can use this resonant circuit to detect the change in surface impedance. As a matter of fact, the kinetic inductance L_s increase leads to a decrease in the resonance frequency:

$$\frac{\delta f}{f_0} = -\frac{\alpha \delta L_s}{2L_s}$$

Here, α is the fraction of the circuit's inductance contributed by L_s :

$$\alpha = \frac{L_s}{L_s + L_g}$$

The resulting frequency shift is small but measurable, allowing for the detection of individual photon events. This phenomenon illustrated on figure 1.2 is at the heart of MKIDs' technology in astronomy.

Some more detailed explanations on the design of the resonators, the advantages and the future development around MKIDs are included in the appendices A, B, C, D and E.

2 Problematics and Objective of the optics project

2.1 Introduction and Historical Context of the project

In the field of observational astrophysics, advancements in detector technology have been instrumental in pushing the boundaries of our understanding of the universe. The MUSIC project (MULTiband Sub/millimeter Inductance Camera) at Caltech was a significant step forward in this regard. MUSIC was a four-band, 14' field-of-view camera, designed to observe sub-millimeter and millimeter wavelengths, and was fielded on the Caltech Sub-millimeter Observatory (CSO) during its last operational years (2012-2015). Despite its groundbreaking technology, MUSIC faced several challenges, including issues with its focal plane's design and manufacturing, which limited its scientific output.

In response to these challenges, the Caltech team initiated a follow-up project called NEW-MUSIC (Next-generation Extended Wavelength-MULTiband Sub/millimeter Inductance Camera). NEW-MUSIC is designed to build upon the successes and lessons learned from MUSIC, with significant innovations aimed at enhancing the instrument's performance. The goal of NEW-MUSIC is to provide extended spectral coverage and sensitivity in the sub-millimeter and millimeter wavelength ranges, making it a crucial tool for studying a variety of astrophysical phenomena, including the cosmic microwave background (CMB), galaxy clusters, and star formation.

The Transition to NEW-MUSIC

The NEW-MUSIC project is characterized by its broad spectral coverage, spanning six bands from 80 to 420 GHz, with polarization capabilities. This instrument is designed to be deployed on the Leighton Chajnantor Telescope (LCT), which will be one of the highest surface-accuracy submillimeter telescopes in the world once operational. NEW-MUSIC aims to address the limitations of its predecessor by employing innovative technologies such as hierarchical, phased arrays of polarization-sensitive superconducting slot dipole antennas, coupled with photo-lithographic band-pass filters, and utilizing a type of kinetic inductance detector (KID) architecture known as MS-PPC-LEKIDs (Microstripline-Coupled Parallel-Plate Capacitor Lumped-Element KIDs).

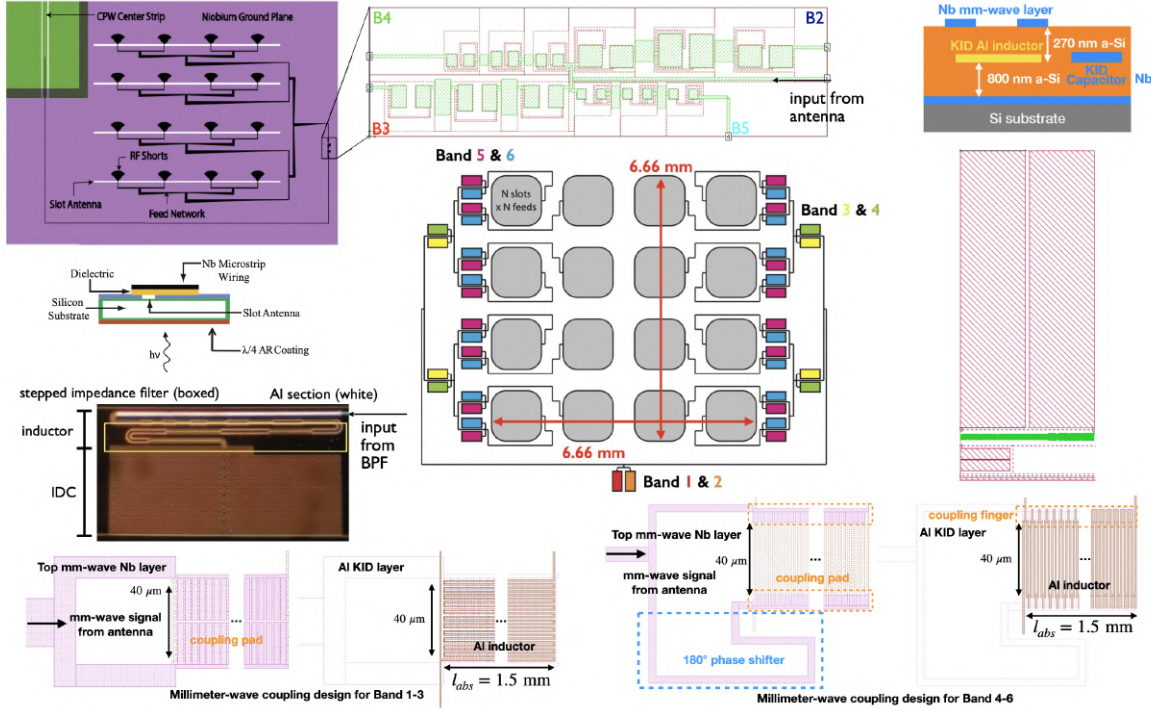


Figure 2.1: MUSIC and NEW-MUSIC design concepts

These innovations are expected to provide background-limited performance and significantly enhance the instrument's flexibility in design and operation. Moreover, NEW-MUSIC will reuse much of the MUSIC instrument's infrastructure, optimizing the scope to a quarter-scale focal plane deployment capable of conducting the majority of NEW-MUSIC's scientific objectives. The design concepts of MUSIC and NEW-MUSIC are illustrated on figure 2.1.

2.2 The Importance of KIDs in NEW-MUSIC

As it is visible in the figure 2.1, at the heart of NEW-MUSIC's technological advancements are the antennas and sub-antennas (the 16 grey square-shaped components on figure 2.1). Each of these sub-antennas is connected to multiple KIDs. As we said KIDs are particularly well-suited for applications in sub-millimeter and millimeter astronomy due to their high sensitivity, ability to operate at very low temperatures, and ease of multiplexing, which allows for the simultaneous readout of thousands of detectors. These characteristics make KIDs ideal for capturing faint and rapidly varying astronomical signals across a wide range of wavelengths.

NEW-MUSIC leverages a specific type of KID architecture known as MS-PPC-LEKIDs (Microstripline-Coupled Parallel-Plate Capacitor Lumped-Element KIDs). This design enhances the flexibility of the detectors and improves their background-limited performance, making them capable of achieving the instrument's ambitious scientific goals. The success of NEW-MUSIC, therefore, heavily relies on the precise characterization and optimization of these antennas connected to the KIDs.

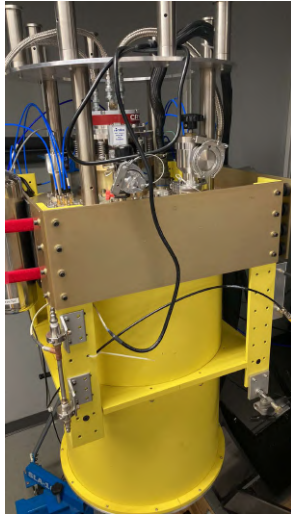
2.3 The Optics Project

Given the critical role that KIDs play in the success of NEW-MUSIC, my project focused on the detailed characterization and optimization of these detectors. During the last testing phases of a generation of detectors, an important step is to be able to associate a KID resonance frequency to its real positions, antennas or sub-antennas pixels, in the detectors array. This is crucial to verify and calibrate the capabilities of the detectors, making it imperative to develop a method for precisely mapping and identifying the detectors outside the cryostat.

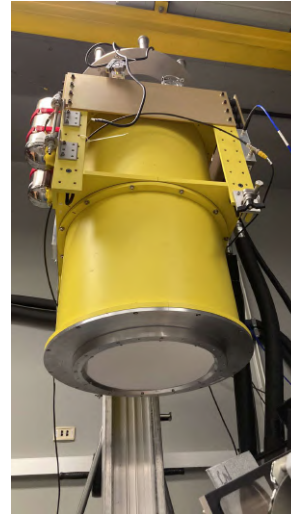
To address this challenge, my work involved constructing an optical setup designed to accurately image the antennas and map their positions. At this point the setup consisted of several key components developed in the next parts.

2.3.1 The Cryostat

At the heart of the NEW-MUSIC system is the cryostat, a specialized cooling device that maintains the detectors at cryogenic temperatures (around 250mK) necessary for superconductivity. The cryostat is crucial because the KIDs must operate at temperatures near absolute zero to exhibit their superconducting properties. More precisely they provide their best working point for a temperature T for which we have $\frac{T}{T_c} < \frac{1}{8}$, T_c being the limit temperature of superconductivity. Inside the cryostat, the detectors are arranged in an array, where each pixel corresponds to some specific KIDs. Some pictures of the cryostat present in the laboratory at Caltech are shown on figure 2.2.



(a) Cryostat up view



(b) Cryostat down view



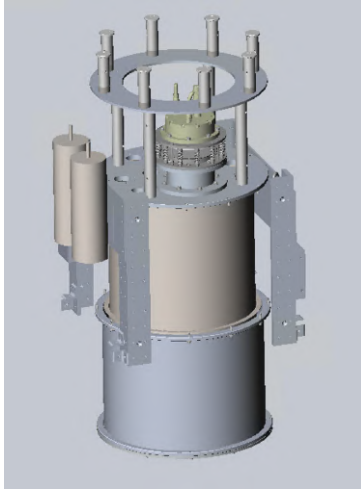
(c) Cryostat left view



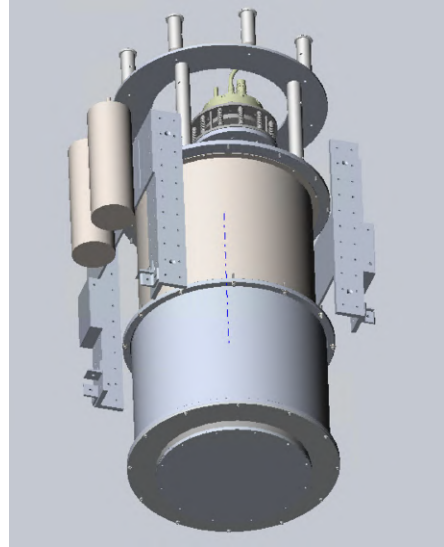
(d) Cryostat right view

Figure 2.2: Photos of the cryostat from different angles

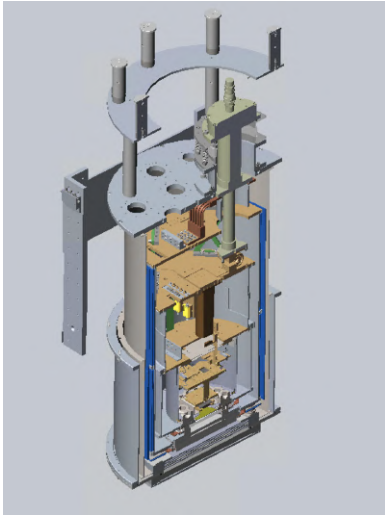
As we said the main role of the cryostat is to cool down the detectors. It isolates them from external thermal noise, ensuring that the signals detected are as pure and unaltered as possible. Our cryostat's design includes multiple thermal and magnetic shields and layers to prevent heat and magnetic fields from influencing the system, which could otherwise disrupt the superconducting state of the KIDs and introduce noise into the measurements. Some 3D model views of the cryostat are shown on the following figure 2.3.



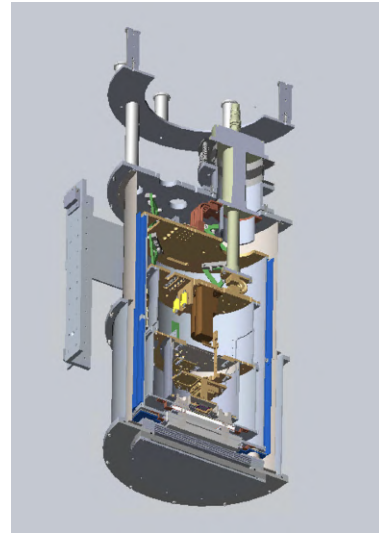
(a) 3D Model of the cryostat with full up view



(b) 3D Model of the cryostat with full down view



(c) 3D Model of the cryostat with cut up view



(d) 3D Model of the cryostat with cut down view

Figure 2.3: 3D Model views of the system from different angles

Finally at one side of the cryostat there is a window letting the light pass from the exterior to the inside where the detectors rely. This configuration concerns the situation where we want to detect signals from the sky and when the system is in real working conditions. Here during the measurements we will consider that a black-body source coming from the beam-mapper (see following subsections) will emit the sub-millimetric signal to the detectors inside the cryostat.

2.3.2 The Detectors

As we said earlier, the important components of the system and which will play a vital role in this project are the detectors. These ones are located inside the cryostat as it is

visible on the figure 2.4 below.

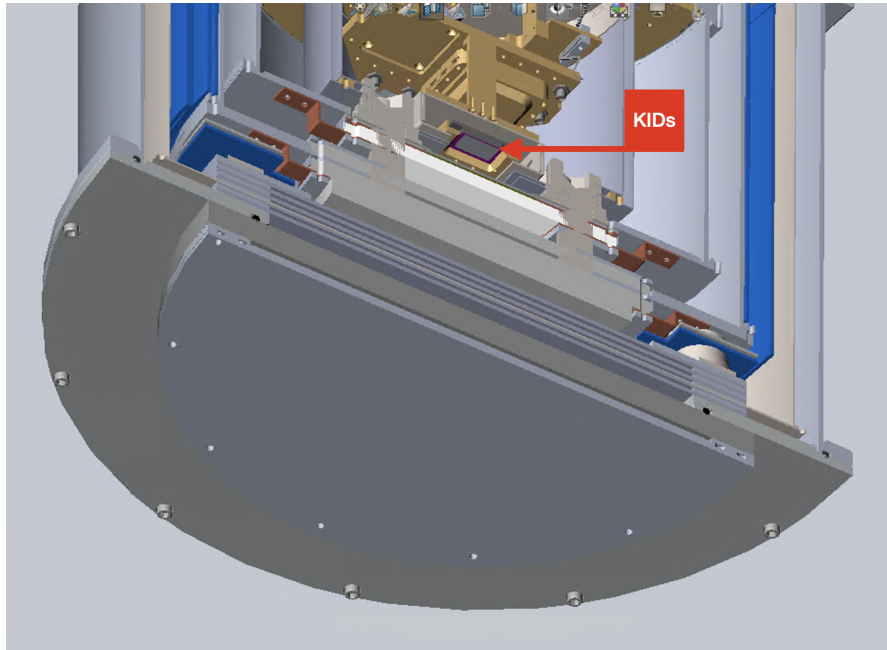
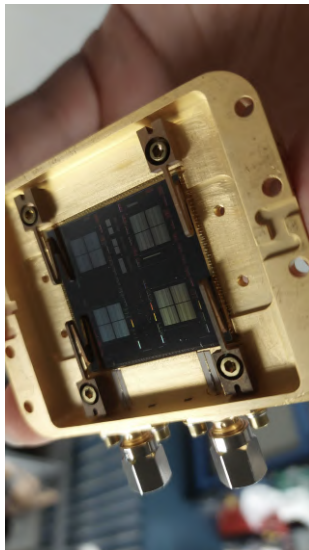
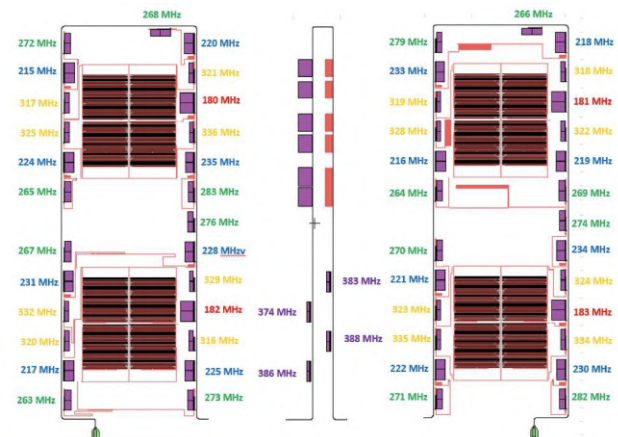


Figure 2.4: Zoomed 3D model view of the detectors inside the cryostat

One should bear in mind that these devices are composed of internal micro circuits:



(a) Picture of one device



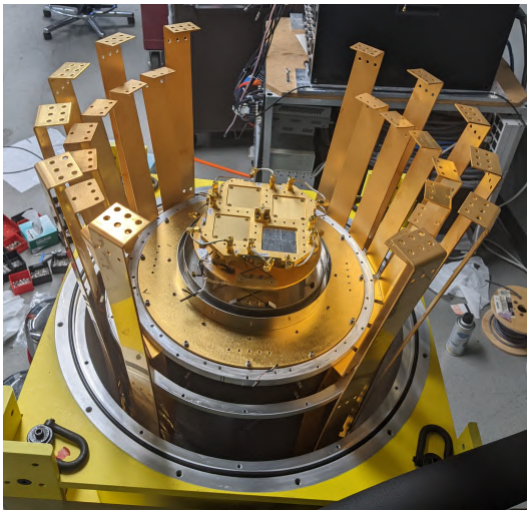
(b) Internal circuits of each device

Figure 2.5: Composition of the devices

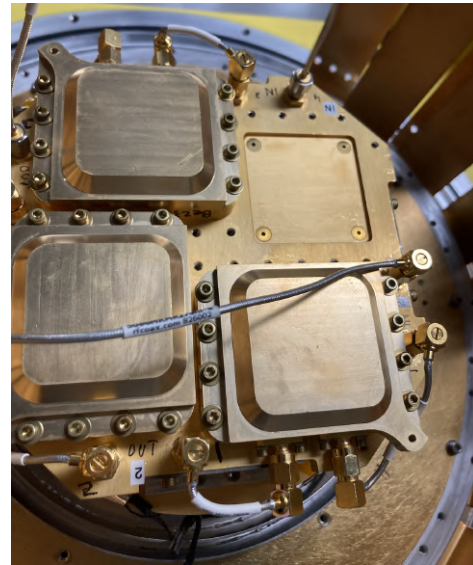
As depicted in these pictures, the internal circuit of the device includes four main squared components, which we will designate as antennas. Each of these antennas is composed of four smaller squared components, which we will refer to as sub-antennas. In total, this configuration results in 16 sub-antennas distributed across the device and all connected to different KIDs.

These sub-antennas are the fundamental detection elements within the system, responsible for capturing the electromagnetic signals that NEW-MUSIC will analyze. Therefore, it is not sufficient to merely recognize and map the four main antennas; the project must extend this capability to the 16 sub-antennas. This finer level of resolution is crucial for ensuring that each individual sub-antenna is correctly aligned, functional, and operating within its specified parameters.

Inside the cryostat, the devices are arranged in a precise array. Each detector or pixel in this array corresponds to a specific location on the focal plane of the instrument. The careful positioning of these detectors is critical because any misplacement can lead to inaccuracies in the data collected, especially in case of misidentification, compromising the overall performance of the NEW-MUSIC instrument.



(a) Picture of the opened cryostat



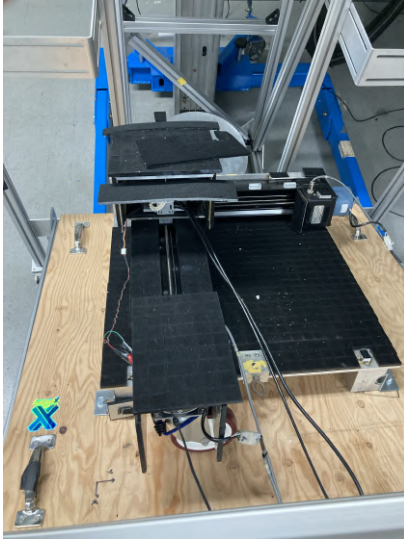
(b) Installation of the detectors

Figure 2.6: Cryostat opening and installation of the devices

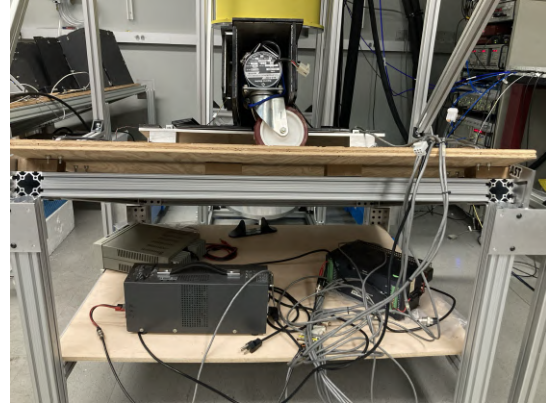
2.3.3 The Beam Mapper

During the development and characterization of a new generation of devices precision and accuracy are essential. That's why the laboratory has a Beam Mapper, a 2D translation stage with an optical source in the form of a black body radiation source and an aperture. This setup plays a critical role in ensuring that the devices are correctly aligned and functioning as intended. Its primary purpose is to track, map and make an image of the beams emitted by the black-body source inside it. The Beam Mapper emits the beams to the detectors according to a detailed spatial input giving information about where each beam lands on the detectors. This mapping process is essential for verifying that the beams are correctly aligned with the detectors and that each detector is capturing the signal as expected. We said earlier that in working conditions the detectors inside the cryostat receive the signal through the window. However during the design phase only, we will consider the reverse optical path from the detectors to the beam-mapper. This is due to the fact that the position of the detectors inside the cryostat is fixed. To properly map the antennas' position we then have to adjust the movable head of the beam-mapper

and place it where the image of the detectors after the optical setup lands on the beam-mapper plane.



(a) Picture of the beam-mapper - up view



(b) Picture of the beam-mapper - down view

Figure 2.7: Pictures of the beam-mapper from different angles

The Beam Mapper allows researchers to make an image from the beams of the antenna used in the detector. In this project, we will use it to correlate the position and intensity of the beams with the physical layout of the antennas inside the cryostat. This correlation is critical for identifying any misalignments or inconsistencies in the positioning of the detectors, which could otherwise lead to errors in the data collected by NEW-MUSIC.

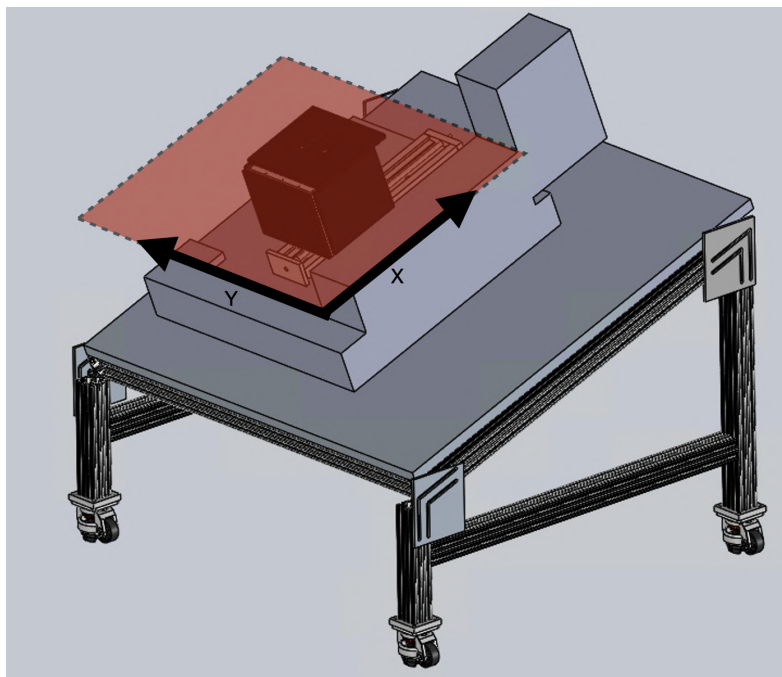


Figure 2.8: 3D model of the beam-mapper with the x and y axis movement of the head

The core functionality of the Beam Mapper is built around two primary components as shown on figure 2.8: an inclined 2D plane and a moving head that contains an aperture and a black body radiation source.

The plane serves as the surface onto which the sub-millimeter wavelength beams are focused. As the beams from the KIDs are emitted, they naturally converge at specific points on this plane. The inclined nature of the plane allows for a broad and accessible surface area where these focal points can be effectively mapped. This design ensures that the entire focal area is covered efficiently, allowing for a comprehensive analysis of the beam profiles.

The moving head is a critical component that carries a precise aperture. This head moves systematically across the 2D plane, allowing to emit signal on a wide area. As the head moves, the detectors inside the cryostat capture the position, intensity, and profile of the beams at various locations, effectively mapping the entire plane. This detailed mapping process enables the Beam Mapper to precisely characterize the beam's properties, including its intensity distribution and spatial coherence.

2.3.4 The objective

To conclude on the context of my project, the primary objective of this project is to address these challenges by developing and implementing an optical setup that allows for the accurate mapping and characterization of the detectors, both at the level of the main antennas and the finer sub-antennas within each detector. Specifically, the project aims to:

- By design, accurately reconstruct the images of the main antennas outside the cryostat: the first goal is to ensure that the four main antennas can be precisely imaged and mapped outside the cryostat and on the beam-mapper plane, allowing for their positions to be correlated with their expected locations within the system.
- Characterize and map each sub-antenna: beyond just mapping the main antennas, the project seeks to achieve a detailed characterization of the 16 sub-antennas. This involves resolving the individual positions of each sub-antenna on the beam-mapper's working plane and assessing their performance to ensure that they are correctly aligned and functioning optimally.
- Optimize setup performance: by identifying any discrepancies or misalignments, the project will enable adjustments that optimize the performance of the optical setup. This is essential for ensuring that the NEW-MUSIC instrument capabilities can be characterized.

In summary, this project is focused on overcoming the critical challenges associated with the alignment and characterization of KIDs within the NEW-MUSIC system. By achieving precise mapping and detailed characterization of the antennas and sub-antennas, the project will characterize the instrument's ability to detect and analyze faint astronomical signals, thereby contributing to the overall success of the NEW-MUSIC initiative.

3 Modelization of the optics setup with Zemax

This part will focus deeper on the design of the optics which will be used to build the image of the detectors outside of the cryostat: an ellipsoidal mirror and a fold mirror. This setup needs to be modeled properly using the optical geometry with minimum tolerance. For this purpose we use a specific software called Zemax to design the system and assess the optical path followed by the light from the detectors in the shape of unidirectional beams. That's why we will first speak about the basic optical phenomenon involved here and then design the optical component needed in this project.

3.1 Optical Systems

In this project we will mainly use the principles behind reflection: The Snell-Descartes 2nd law of reflection.

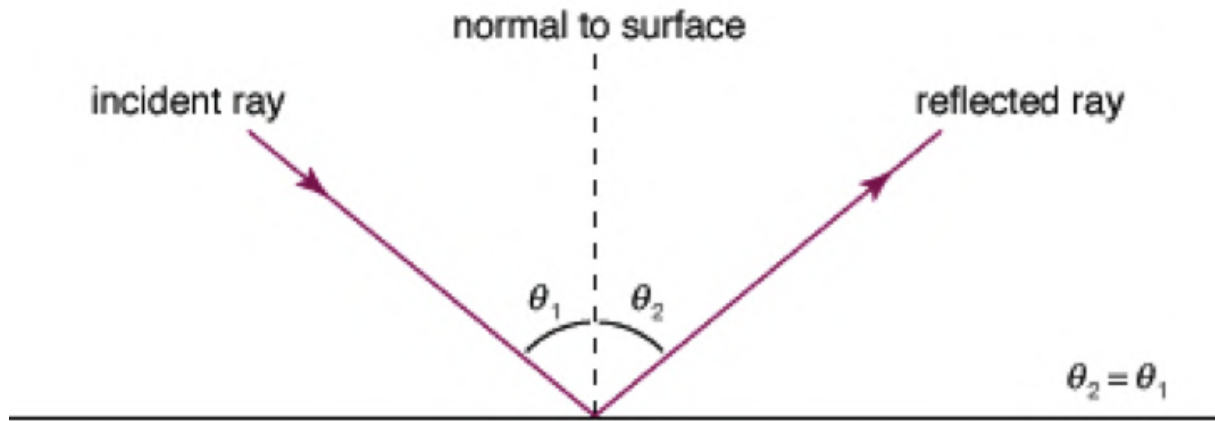


Figure 3.1: Illustration of the principle of reflection [6]

This law states that when light rays encounter a surface and bounce back into the medium from which they originated, the angle of incidence (the angle between the incoming ray and the normal to the surface) is equal to the angle of reflection (the angle between the reflected ray and the normal). On figure 3.1 we get $\theta_1 = \theta_2$. Mirrors utilize this principle to redirect light. These are the main optical components we will use in this project. These components will be developed in the following subsections.

3.1.1 Ellipsoidal Mirrors

Ellipsoidal mirrors are a type of curved mirror shaped like a part of an ellipsoid. They have two focal points, and their unique geometry allows them to focus light from one focal point to the other. This property makes them invaluable in correcting off-axis aberrations, a common issue in imaging systems where light from an object point not on the optical axis is focused. [7]

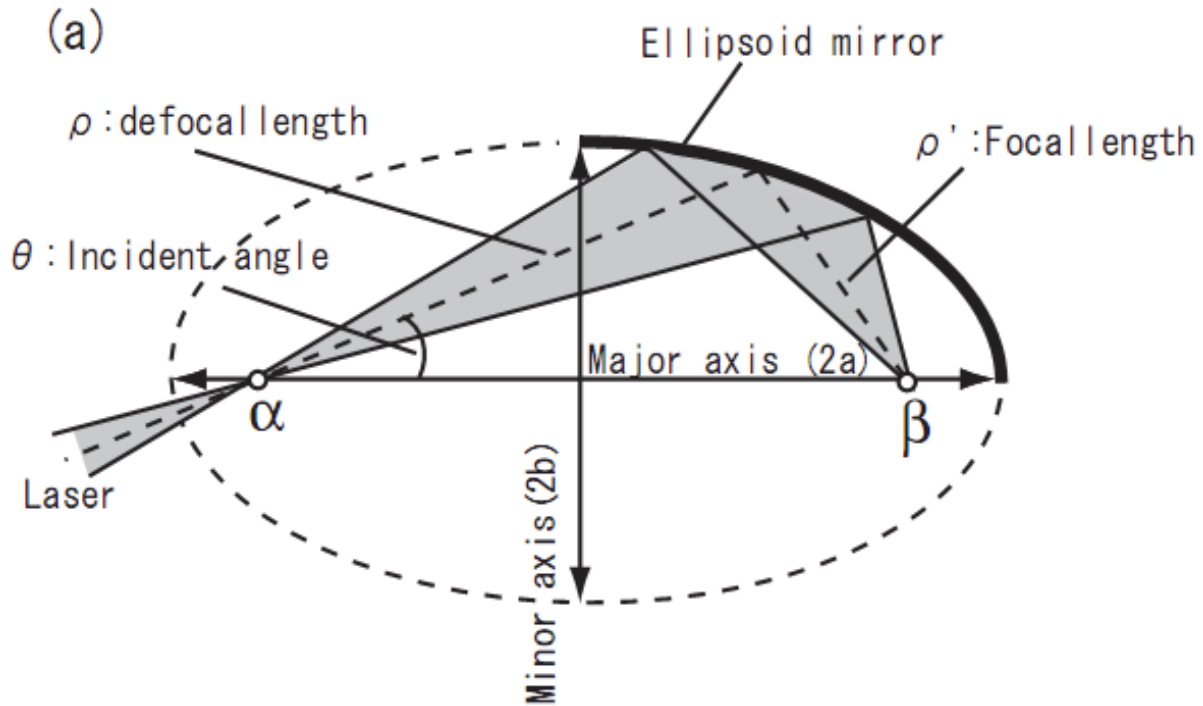


Figure 3.2: Schematics of an ellipsoid and its parameters [7]

Let's focus on the design principle:

- **Geometry:** The shape of an ellipsoidal mirror is derived from an ellipsoid, defined as the range of points where the sum of the distances from two fixed points (the foci) is constant. When designing an ellipsoidal mirror, the positions of these foci are critical, as they determine where the mirror will focus incoming light.
- **Optical Path:** In an optical system using an ellipsoidal mirror, rays emanating from one focus will reflect off the mirror surface and converge at the second focus. This property is leveraged to design compact and efficient optical systems that require light to be focused or re-directed with minimal aberration.
- **Aberration Correction:** Ellipsoidal mirrors are particularly effective in correcting spherical aberrations and coma for off-axis light paths, enhancing the quality of the image produced by an optical system.

In our project they could be a particularly good fit as they make it possible to refocus the beams coming from one of their foci to the other. It means that if we put the detectors to characterize at one of the focal points of an ellipsoidal mirror, this latter can rebuild the image on its second focal point. It is this principle that we will use in this project.

3.1.2 Fold Mirrors

Fold mirrors, also known as folding mirrors, are used to change the direction of light paths within an optical system without altering the image's orientation or quality. They are flat mirrors that introduce no additional aberrations, making them ideal for compacting the

design of optical systems or redirecting light paths for integration into confined spaces. [8]

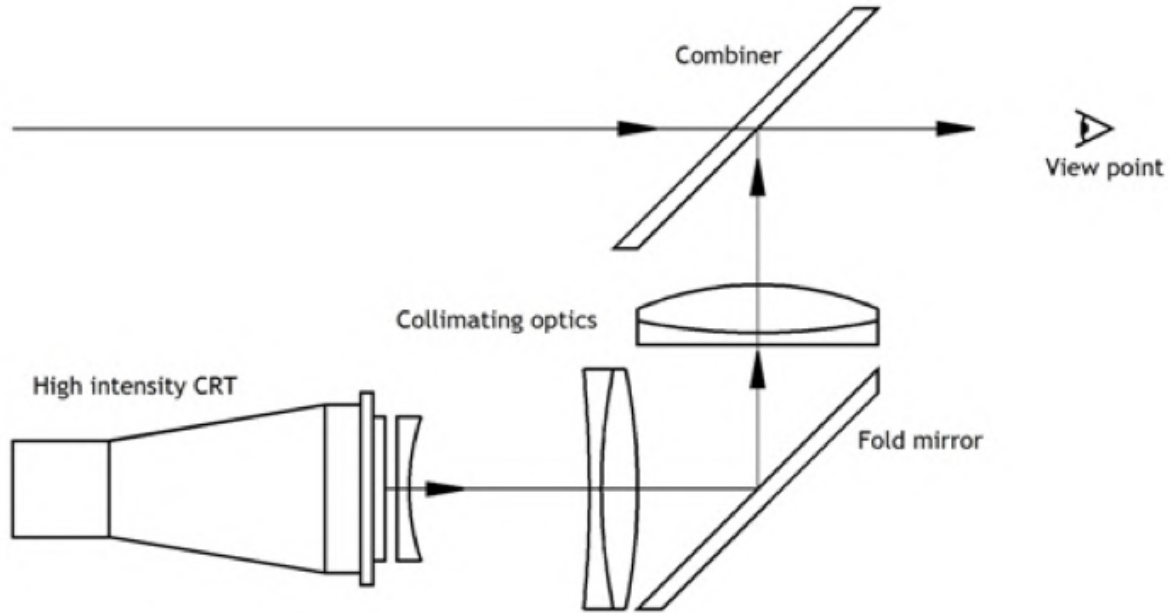


Figure 3.3: Example of application of a fold mirror to redirect light [8]

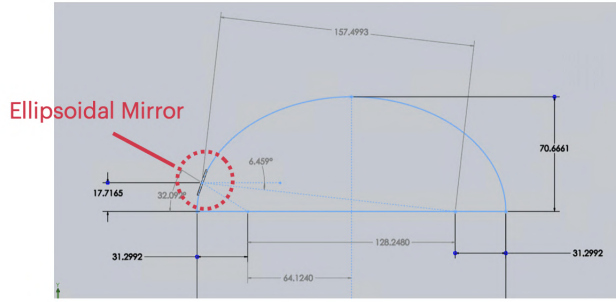
Their ability to redirect the light coming from an object source to any chosen direction (depending on the angle between the beam and the surface of the mirror) make them ideal for our project. Indeed, as the cryostat is directed to the ground, the ellipsoidal mirror can redirect the beams in an ascending direction. We will then need to redirect the beams toward the beam mapper, hence the necessary use of a fold mirror.

3.2 Setting Up Zemax for the project

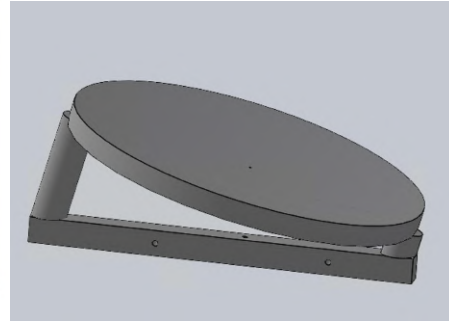
In this part we will focus on the modelization of the setup on Zemax. As said earlier this setup will be composed of the detectors (inside the cryostat), the ellipsoidal mirror, the fold mirror and finally the mapping plane of the beam-mapper. Also, the wavelength of the signal for which the detectors are designed is around 0.6mm (sub-millimeter applications).

3.2.1 Characterization of the Ellipsoidal Mirror

The first step here was to characterize the ellipsoidal mirror needed in our project. The mirror that will be used is an ellipsoidal mirror provided by NASA's Jet Propulsion Laboratory. By definition it is a mirror built out of an ellipse with the following CAD and dimensions also provided by JPL:



(a) Ellipsoidal mirror's ellipse of construction (dimensions in inches)



(b) 3D model of the Ellipsoidal mirror

Figure 3.4: Model of the ellipsoidal mirror provided by NASA's JPL

We can deduce from the ellipse used to create the mirror on figure 3.4 the standard parameters of the ellipse: $a = 95.42$ inches, $b = 70.66$ inches. By using the equation of the eccentricity we get:

$$e = \frac{\sqrt{a^2 - b^2}}{a} = 0.6720 \quad (1)$$

When transitioning from the conceptual and visual domain of CAD to the precise, simulation-driven environment of Zemax, a direct application of these familiar parameters (a , b , e) is not feasible. Zemax approaches the modeling of ellipsoidal mirrors differently. It utilizes a set of parameters optimized for optical simulation and analysis, specifically the conic constant (K) and the radius of curvature (R) of the mirror. This discrepancy between CAD and Zemax parameterization necessitates a conversion process, ensuring that the physical mirror's properties are accurately represented in the simulation environment. For that we need to convert the geometric parameters of the ellipse into parameters computed on Zemax. For that we will follow the conversion process detailed in the Appendix F.

Thanks to this conversion process we get the following values:

Parameters ellipsoid			Zemax input parameters	
Semimajor axis a	Semiminor axis b	Eccentricity e	Radius of curvature	Conic Constant
95.42 inches	70.66 inches	0.6720	54.643 inches	-0.823

By computing the values of K and R we can modelize the mirror and then verify by putting the source at one of its foci if the beams are focused at the second one:

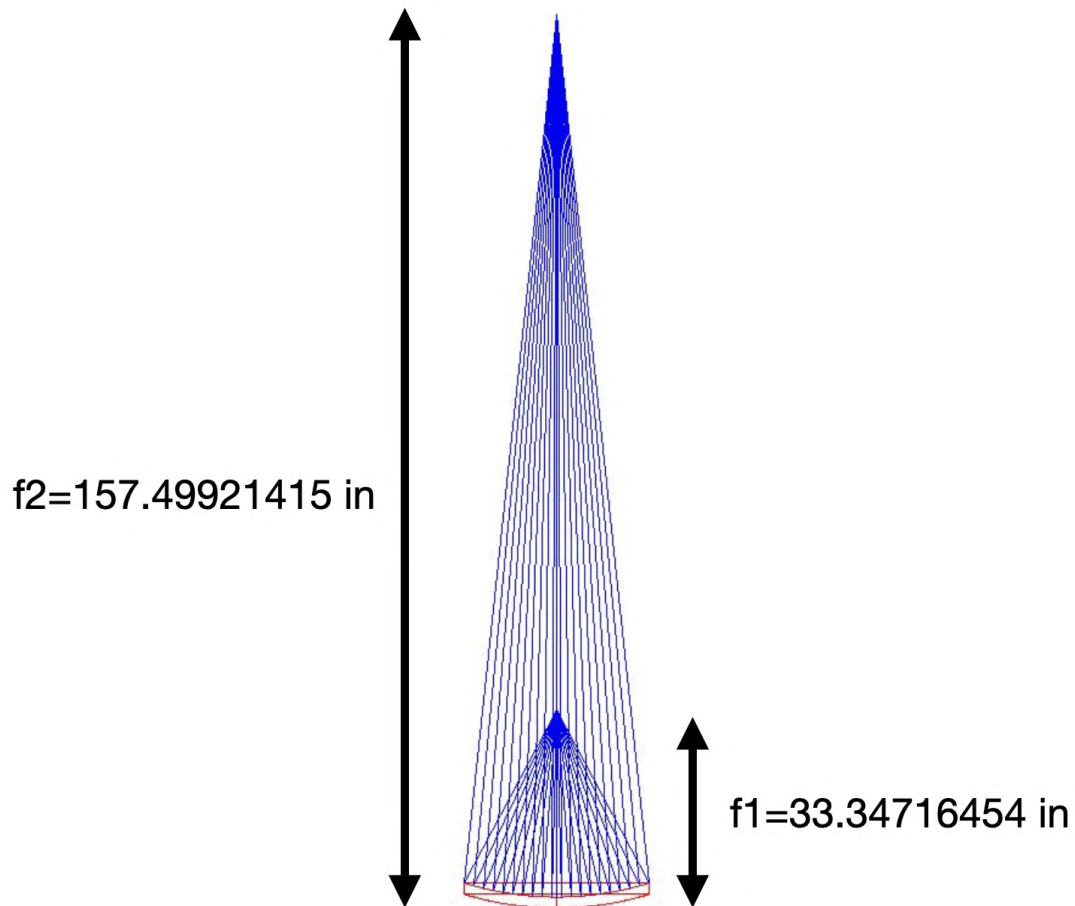


Figure 3.5: Zemax model of the ellipsoidal mirror

As shown on figure 3.5 we find $f_1 \approx 33.347in$ and $f_2 \approx 157.4992in$ which are really close to the one we found on the CAD model. We can even overlay the Zemax model and the ellipse of construction to see that the beams coming from one of the foci converge to the second as shown on the following figure 3.6:

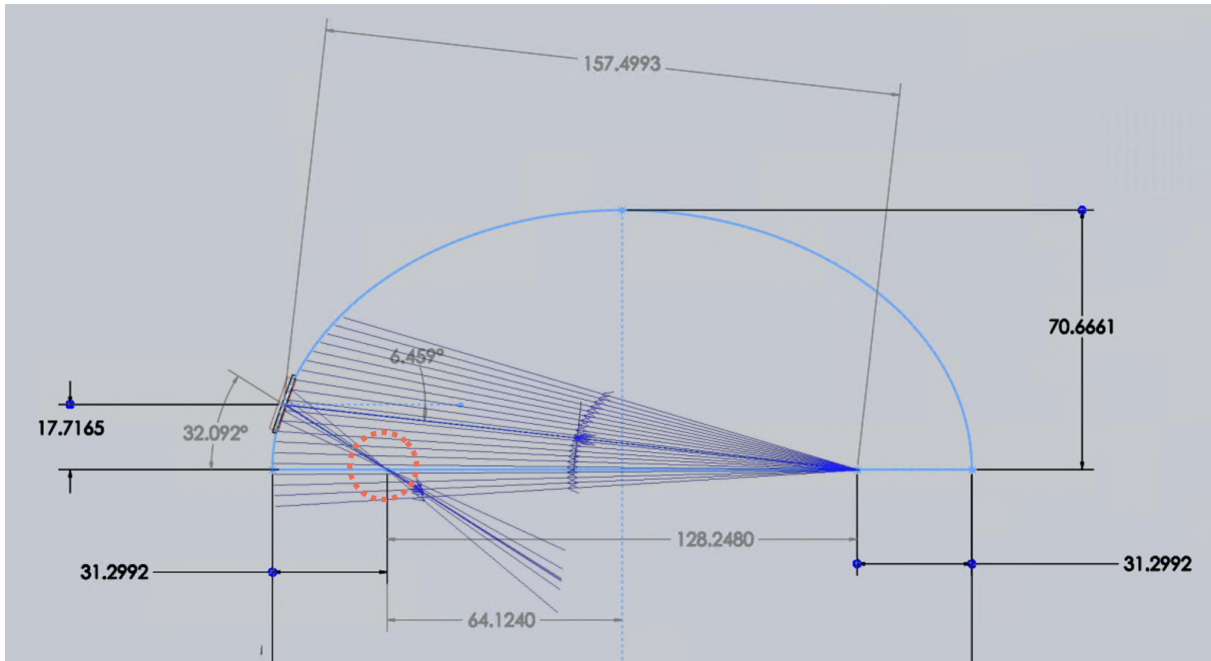


Figure 3.6: Overlay of Zemax model with 3D model

3.2.2 Implementation of the whole structure

With the model of our ellipsoidal correctly implemented on Zemax, we can build the entire model. For that we will take into account the size of the window as shown on the CAD model picture of figure 3.7. We will also use a cone-angle aperture for our object (the detectors) with a value of 40 degrees which corresponds to the angle of deviation of our beams coming out of the cryostat's window.

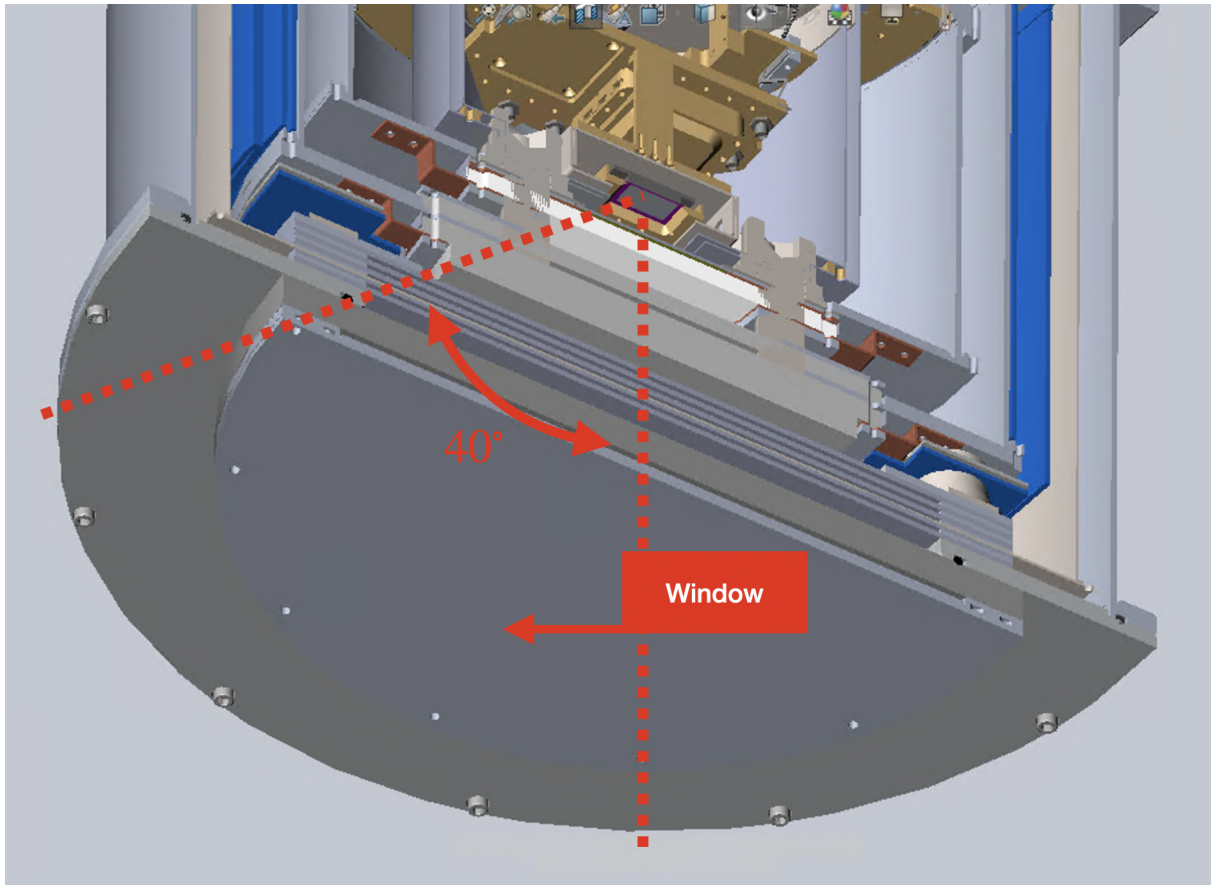


Figure 3.7: 3D Model of the cryostat with its window

After setting up all the structure on Zemax we get on the following figure 3.8 a first simulation with the complete optical path:

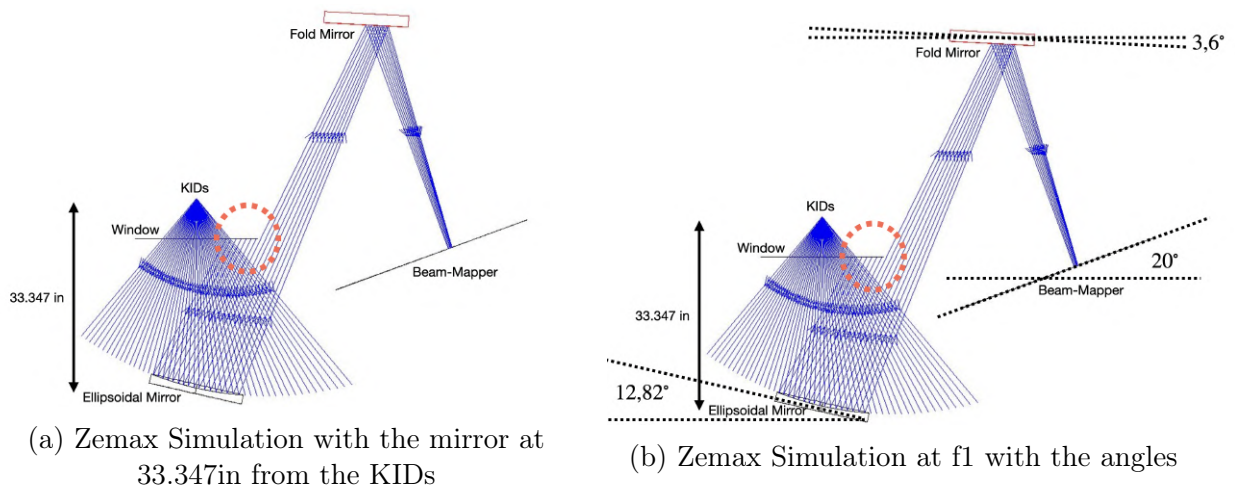


Figure 3.8: Zemax Simulation at f1

As we can see on the red circles placed on figure 3.8, nearly half of the beam coming out of the ellipsoidal mirror is blocked by the cryostat's bottom edges. In can cause problems in the beam Gaussian repartition of the energy. This can be verified by simulating the Airy disk on our image beam as shown on figure 3.9:

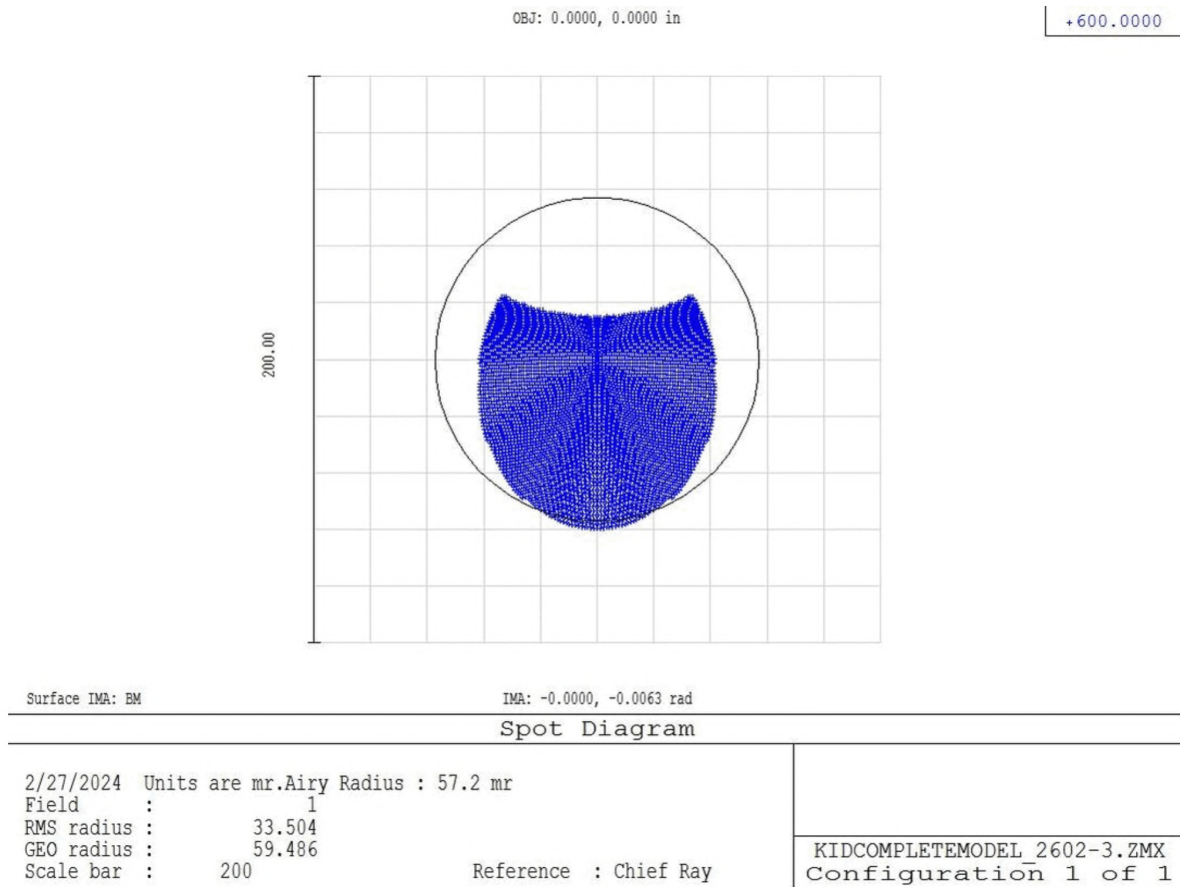


Figure 3.9: Picture of the simulated Airy Disk with part of the beam cut by the window

As we can notice on figure 3.9 a fairly large portion of the beam will be cut by the edge of the cryostat. This is even more an issue as this simulation shows a centered device in the cryostat. In reality the 3.3mm object source on this model represents one sub-antenna. As there are 4 sub-antennas along the horizontal axis the beams can move up to 6.6mm on the left or on the right which can cause even more beam obstruction at the edge. We will then decide to increase the distance between the detectors and the ellipsoidal mirror. By voluntarily working at a distance farther than the first focal length we may accommodate a better image reconstruction on the beam-mapper as less part of the beams will be obstructed by the cryostat. One should bear in mind that it won't cause any deterioration of the image because of the focus on the Beam-mapper as we are working in the sub-millimetric field. However it will engender a change in the orientation of the image as we will see in further sections. Finally by taking a distance KIDs - Ellipsoidal mirror of 40in instead of 33.347in we get the following results presented on figure 3.10:

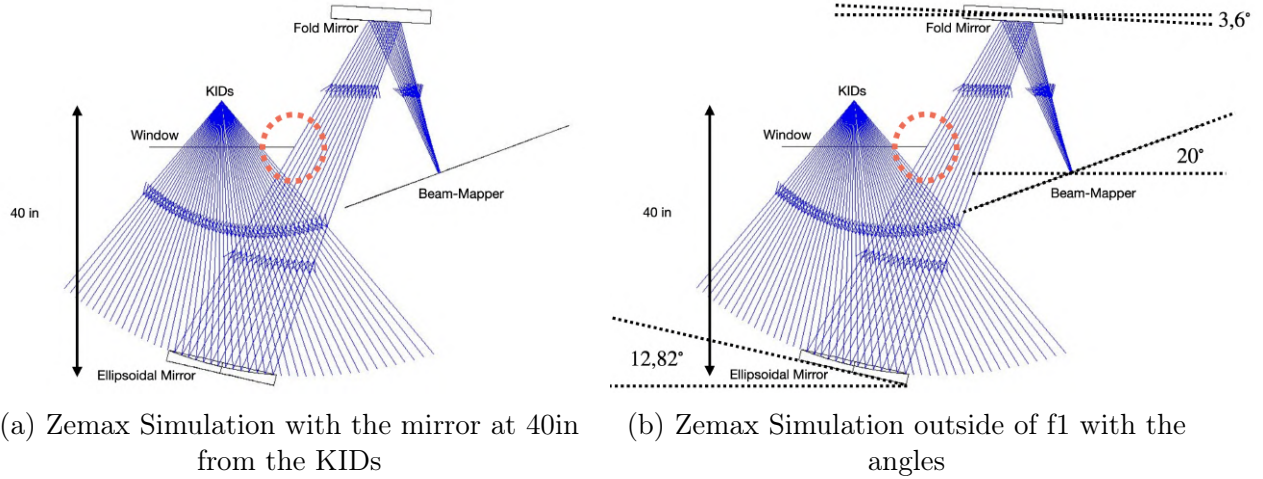


Figure 3.10: Zemax Simulation outside of f1

We can clearly see on the red circles placed on figure 3.10 the impact on the beam obstruction. Almost all the beam can be reflected to the fold mirror when the antenna is centered in the cryostat. Let's see how the Airy Disc is impacted when we move the antenna from -6.6mm to +6.6mm on the horizontal axis inside the cryostat (these values correspond to the distances from the center of the detector to the left and right edges of the device shown on figure 2.5) :

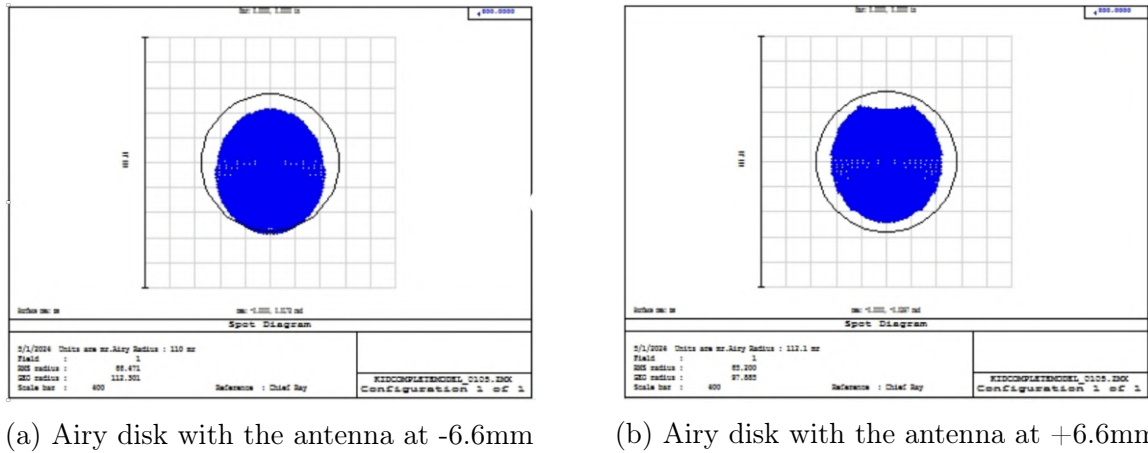


Figure 3.11: Influence of the position of the antennas at -6.6mm and +6.6mm on the simulation of the Airy Disk

The two airy disks present on figure 3.11 show that if the device is far left in the cryostat (position -6.6mm), the disk is complete but its bottom is slightly out of the circle implying that the beam is not obstructed but we are slightly out of the refraction limit, and if the device is far right in the cryostat (position +6.6mm), the disk is completely inside the circle so no refraction but a slight top part of the beam will be obstructed. Then even in the worst case we can ensure that almost all of the beam is reflected to the fold mirror. The height of the fold was adjusted to ensure the beams are converging on the beam-mapper's plane. As we can see this height is more important if we put the ellipsoidal mirror on f1. Putting it at 40 inches from the detectors makes it possible to have the fold mirror closer. This then concludes the design of the optics setup.

4 Mechanical design of the structure

In the previous we could design the optics setup with the proper distances and orientations between all the components. However, we need to build the setup. This part will then focus on the CAD design of the structure which will hold the optics and on the assembly of the whole structures in the laboratory at Caltech.

4.1 Choices of design

As said earlier, there are 4 main components in the optics setup:

- The cryostat with the detectors
- The Ellipsoidal Mirror
- The Fold Mirror
- The Beam-Mapper

The cryostat and the beam-mapper already have each independant structures composed of standard 8020 aluminium frames. The idea would then be to build a structure that would hold all the components together. Building a structure for each of the mirror and having 4 independent structure could engender misalignment problems and increase the time of implementation due to all the distances to precisely ensure between the components. Then I decided to build two separate structures:

- One structure coupling the ellipsoidal mirror to the cryostat and detectors : this is mainly due to the convenient presence of the ellipsoidal mirror just below the cryostat's window and the fact that they are concentrically aligned
- One structure coupling the fold mirror to the beam-mapper: this is mainly due to the convenient presence of the fold mirror just above the beam-mapper

There are then two structure to design. These two structures will be independent but can be attached in a precise way to ensure global alignment during measures. Also, to ensure a proper and easy coupling compatibility these two structures will be designed out of standard 8020 elements.

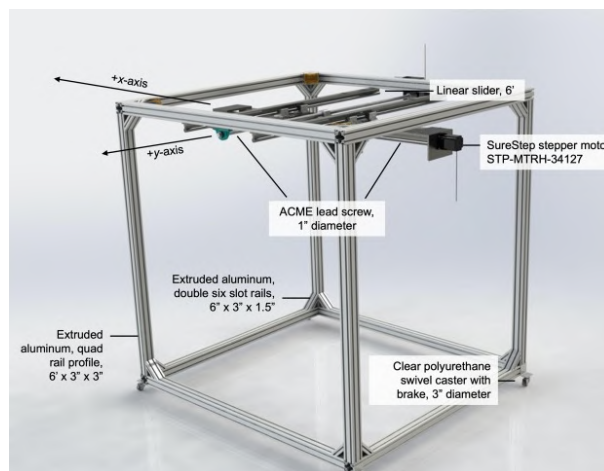


Figure 4.1: Example of structure composed of McMaster Carr and 8020 components

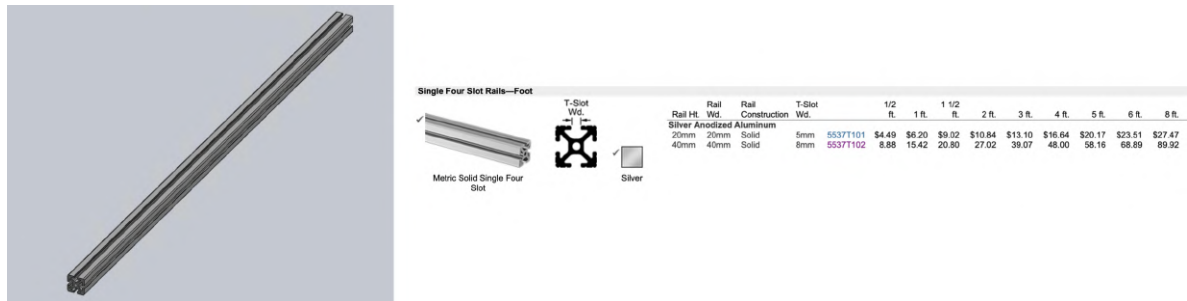
In the following subsections I will then develop the design of the two structures.

4.2 Design of the structures on Solidworks

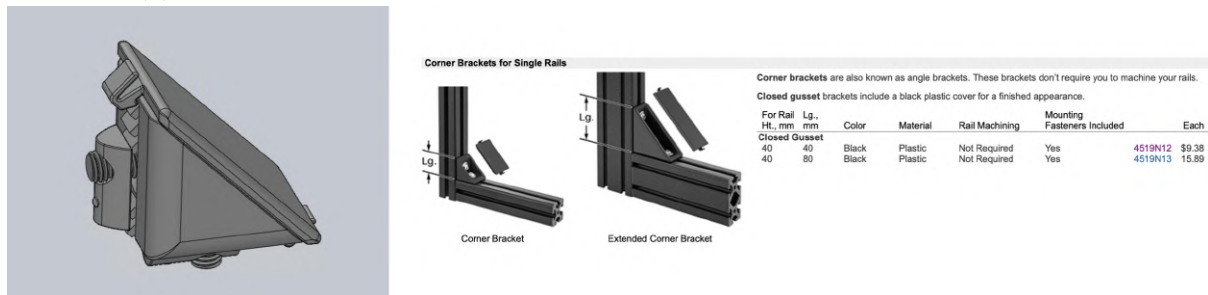
For the design we will mainly use Solidworks as most of the CAD models provided by constructors are made for Solidworks.

4.2.1 Structure Cryostat / Ellipsoidal Mirror

The idea here was to build a structure able to be attached on the four corners of the cryostat. This configuration allows the ellipsoidal mirror to be concentrically aligned to the window of the cryostat and then to the detectors. The structure was made out of two framed rectangles linked by vertical bars. The main elements of the structure are displayed below:



(a) McMaster Carr frames



(b) McMaster Carr angles

Figure 4.2: Main component of the structure

These components are made out of aluminium. This has the advantage of being pretty light but still resistant enough to ensure mechanical resistance.

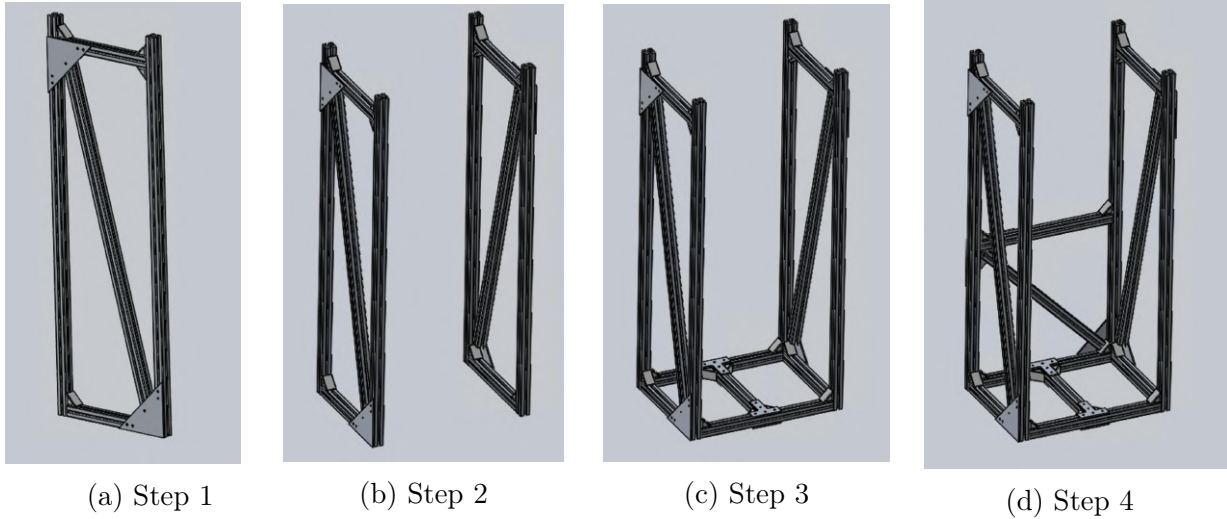


Figure 4.3: Steps of construction of the ellipsoidal mirror structure

The design of one side of the structure is displayed on the step 1 of the figure 4.3. The aluminium triangles and the transversal frames were put in order to add more rigidity to the whole assembly. After designing the other side we get the assembly as shown in the step 2. We can couple these two sides with a T-shaped interface as shown in the step 3. This interface will also serve to attach the ellipsoidal mirror. Finally, as we saw earlier the beam crosses only one side of the structure which means we can use the same method as before to add rigidity on the back of the structure as shown in the step 4.

To attach the ellipsoidal mirror we need to add an interface bar which make an intermediate between the structure and the mirror itself. This is due to the fact that the mirror cannot be modified especially in terms of screw holes and dimensions. This attach and the design of the interface are shown on the figure 4.4

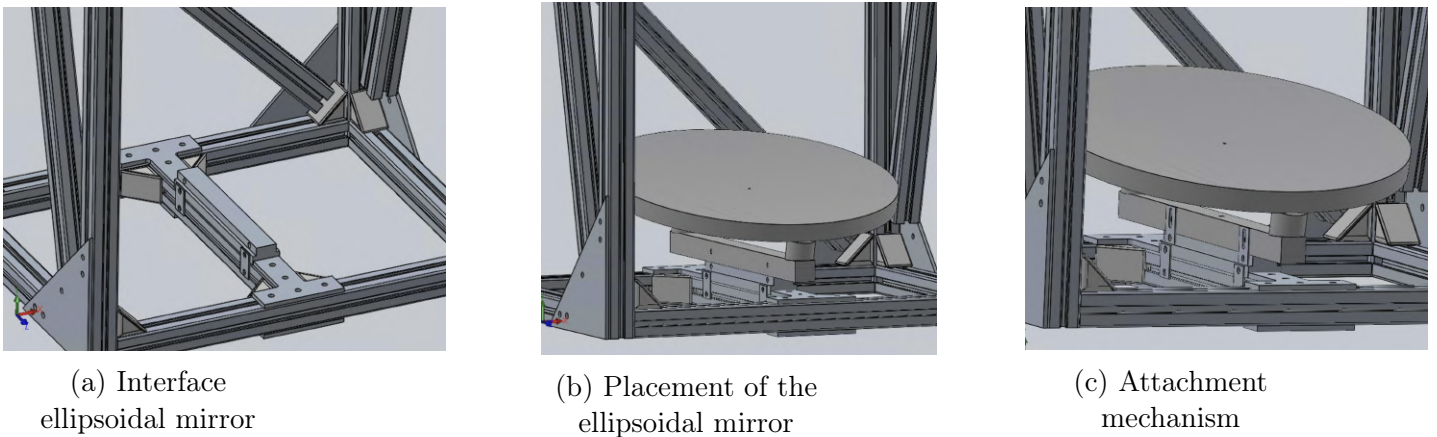
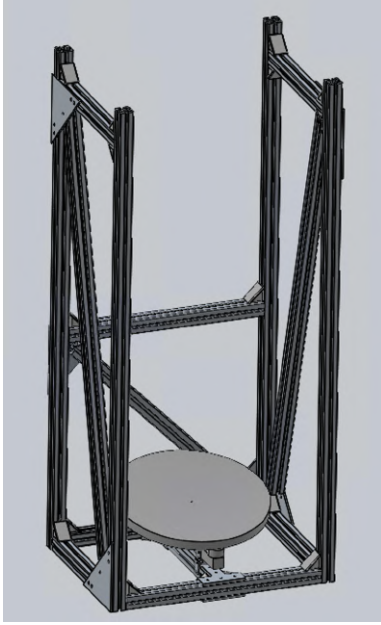
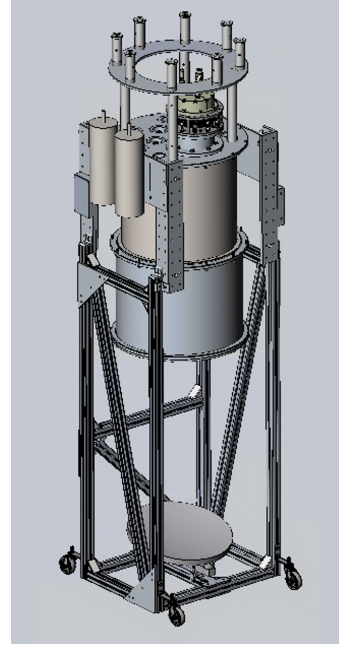


Figure 4.4: Design of the interface between the ellipsoidal mirror and the structure

We finally get the complete assembly which can be then attached to the cryostat. The whole assembly is shown below on the figure 4.5



(a) CAD Model of the final ellispoidal mirror structure



(b) CAD Model of the final ellispoidal mirror and cryostat assembly

Figure 4.5: CAD models of the whole assembly

To conclude the design of this structure wheels have been added to the 4 corners of the ellispoidal structure to ensure we can move easily the structure when attaching it to the cryostat and decrease the mechanical constraints on the frames. Finally as shown on the figure 4.6 we can measure on Solidworks the distance between the KIDs and the ellispoidal mirror and then verify it is 40 inches just like on the Zemax model.

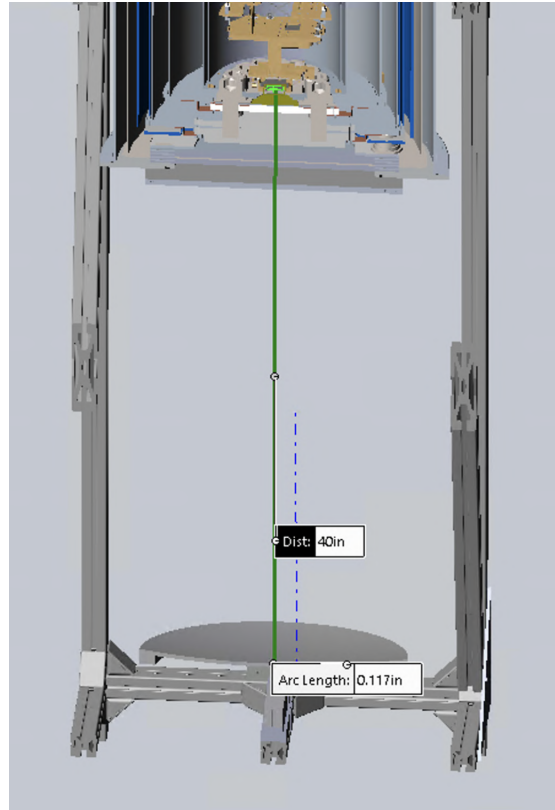


Figure 4.6: Verification distance KIDs - Ellipsoidal Mirror

4.2.2 Structure Beam-Mapper / Fold Mirror

In this part I will detail a bit less the design as the structure holding the fold mirror and attached to the beam-mapper is globally inspired from the design developed in the previous part. The dimensions are different in order to accommodate the presence of the beam-mapper but the method used to design are the same as previously. We base structure for the fold mirror is shown below on figure 4.7:



Figure 4.7: CAD model of the fold mirror structure

However we need a specific interface to hold the fold mirror. Assuming that this latter will be a basic plate we can create 4 tapped holes on the mirror and fix this one to a frame which will be coupled to the structure thanks to a pivot. The result is shown on figure 4.8.

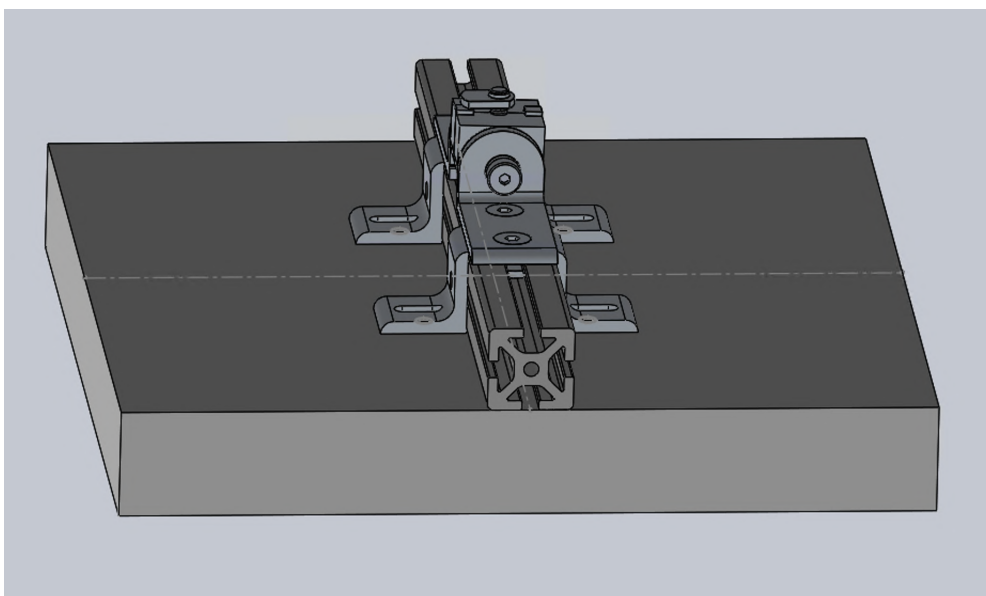


Figure 4.8: Attachment and interface of the fold mirror

I also attached wheels to the corners of the structure for the same purpose as earlier. The final fold mirror assembly is shown on figure 4.9.



Figure 4.9: Final fold mirror assembly

In order to attach it to the beam-mapper structure I designed four plates working as an interface between the aluminium frames of the beam-mapper and that of the fold mirror structure. These plates are shown the following figure 4.10.

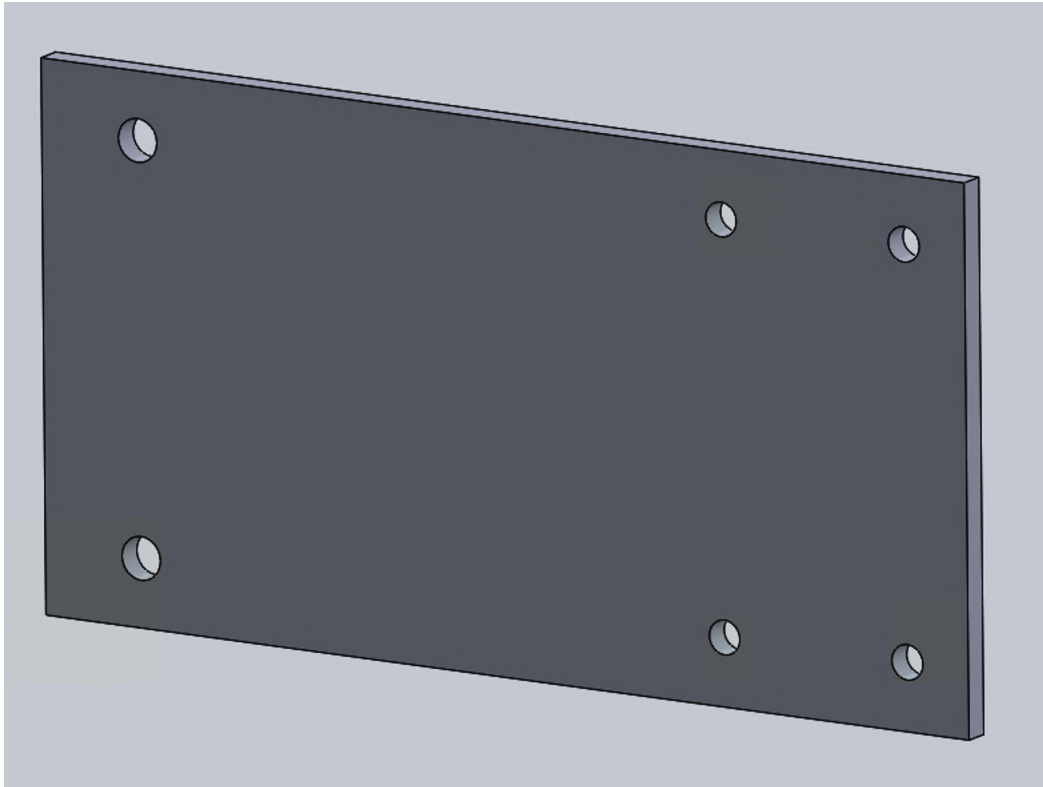


Figure 4.10: Coupling plate used to attach the fold mirror structure to the beam-mapper structure

We finally get the whole fold mirror and beam-mapper attached together as shown on figure 4.11 below.

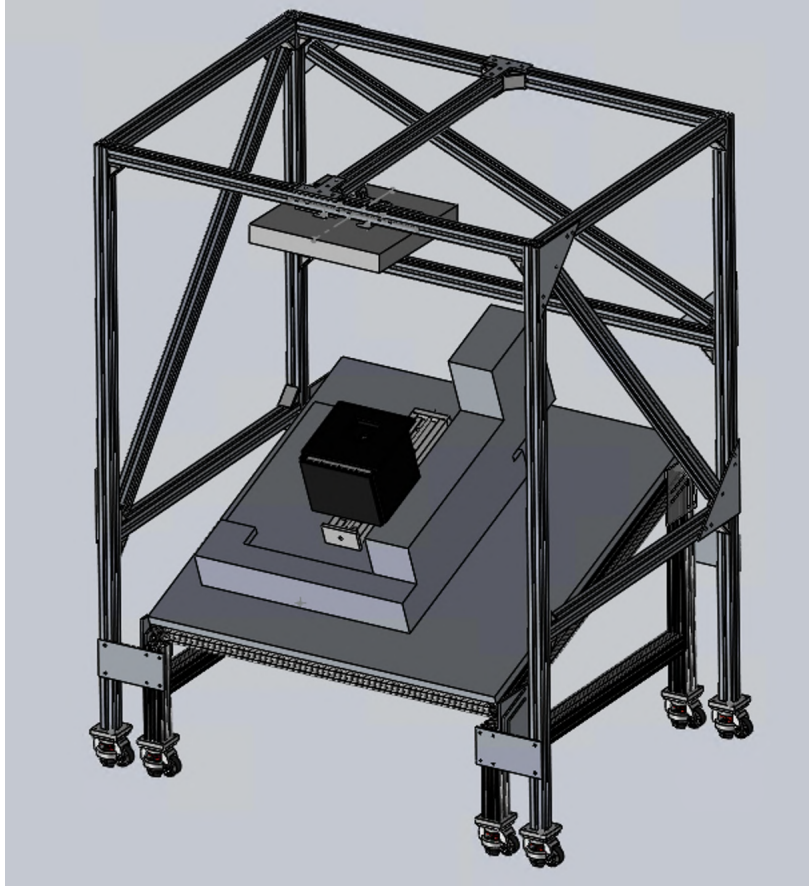


Figure 4.11: Final beam-mapper and fold mirror assembly

After some calculations on the weight of the structure I decided to change the model of wheels used. The ones visible on figure 4.11 is more resistant to the weight and has a modifiable distance to the ground which makes it possible to easily align the structure to the rest of the setup.

4.2.3 Final assembly

Now we have the two main structures we have to design a proper interface making it possible to align them and fix the distance and orientation between them. I chose to design a L-Shaped aluminium plate as displayed below on figure 4.12:

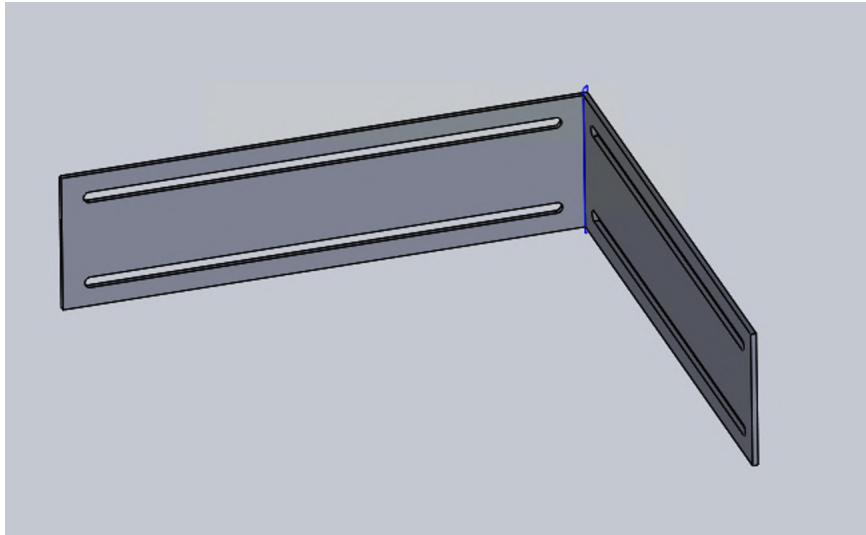


Figure 4.12: L-shaped aluminium interface between the two assemblies

We can finally attach the two assembly and get the whole structure with the cryostat, the ellipsoidal mirror, the fold mirror and the beam mapper, as shown on figure 4.13:



Figure 4.13: Final complete assembly

To verify the coherence of our design as regard to the simulation performed in the previous section, we can overlay our Zemax model on the CAD model to see that all the

dimensions are perfectly respected as we can on figure 4.14.

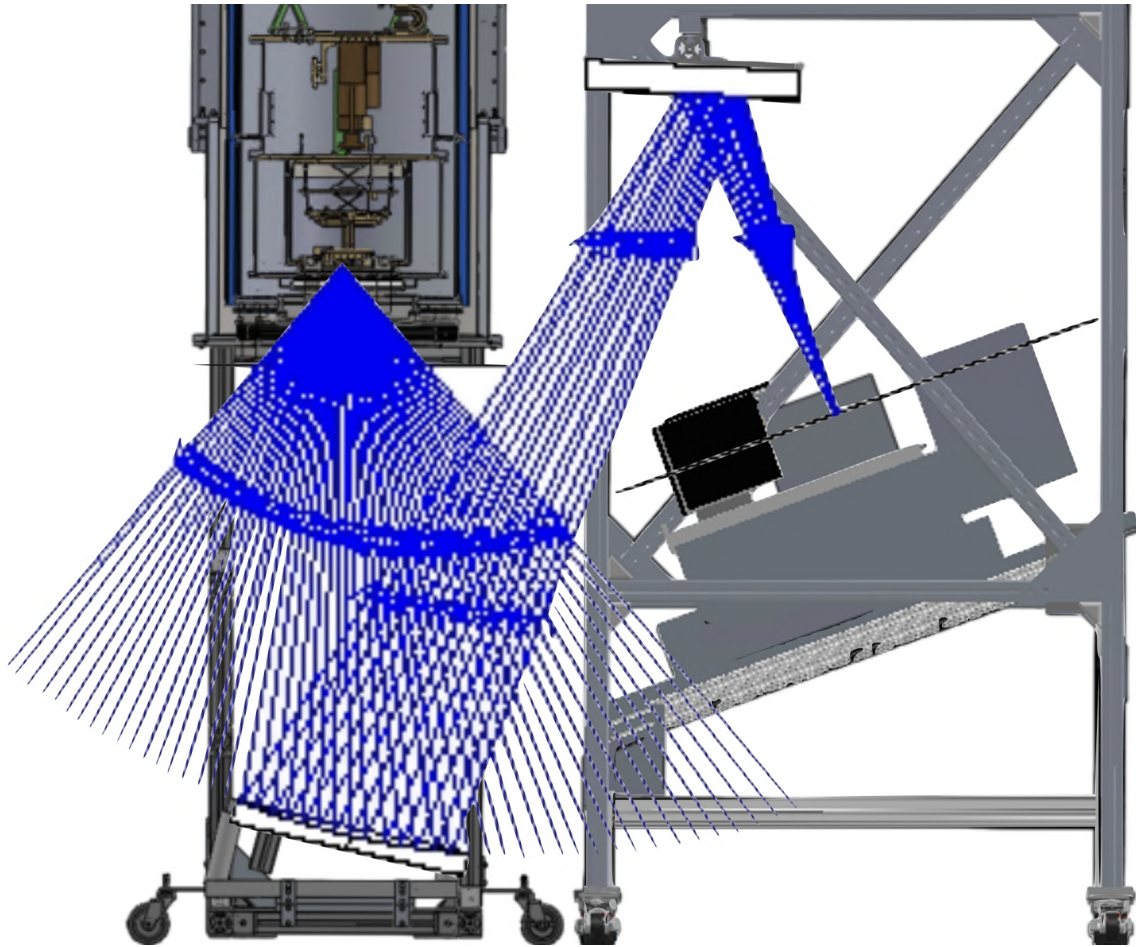


Figure 4.14: Overlay of the Zemax model on the complete assembly CAD model

In the following part we will focus on the fabrication and the building of the whole assembly in the laboratory at Caltech.

5 Construction of the assembly and measurements

This part will serve as a conclusion for this project. We will delve in the construction of the whole assembly and then more importantly in the measurements that have been made and the characterization made possible thanks to this project.

5.1 Construction of the whole assembly

All the components of the assembly were ordered from five different providers:

- Ellipsoidal mirror: provided by NASA's Jet Propulsion Laboratory
- Frames, wheels, pivot, screws and washers: provided by McMaster Carr and 8020
- Fold mirror: an aluminium MIC6 plate with specific surface properties (flatness and smoothness) provided by McMaster Carr
- Custom designed aluminum parts: provided by Xometry

The process to build the structures was the same as the one to design them: building first the sides, attach them with the frames, consolidate the back and sides with lateral frames and the building the interfaces and attach the mirrors. The figures 5.1 and 5.2 shows the first steps of the construction and the fit check performed on the cryostat.

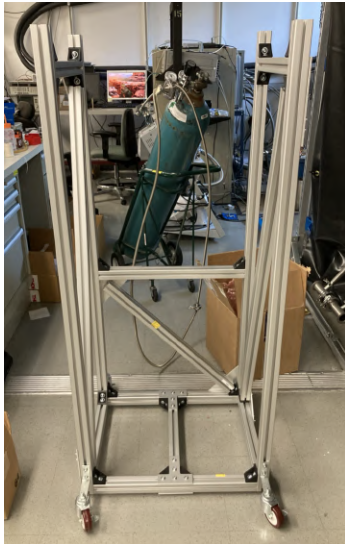


(a) Two sides of the ellipsoidal structure



(b) Intermediate frames used to attach the sides together

Figure 5.1: First steps of building the ellipsoidal structure



(a) Picture of the consolidated structure with the wheels



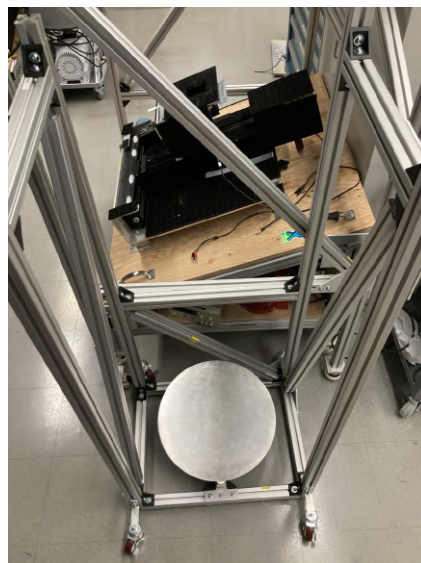
(b) Picture of the fit check between the structure and the cryostat

Figure 5.2: Consolidation and fit check on the cryostat

We finally obtain the final ellipsoidal mirror structure as shown on figure 5.3:



(a) Left side view
ellipsoidal mirror
structure



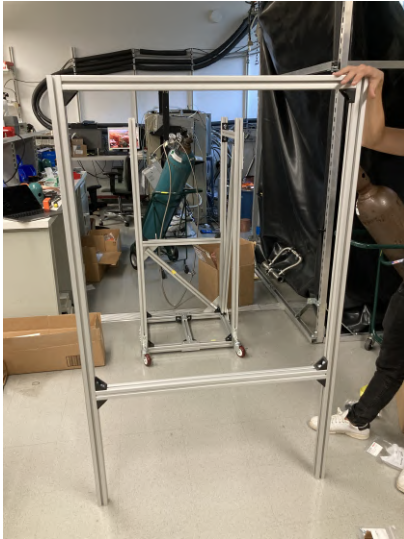
(b) Front view
ellipsoidal mirror
structure



(c) Right side view
ellipsoidal mirror
structure

Figure 5.3: Views of the ellipsoidal mirror structure from different angles

We then do the same with the fold mirror structure, as shown on figure 5.4:



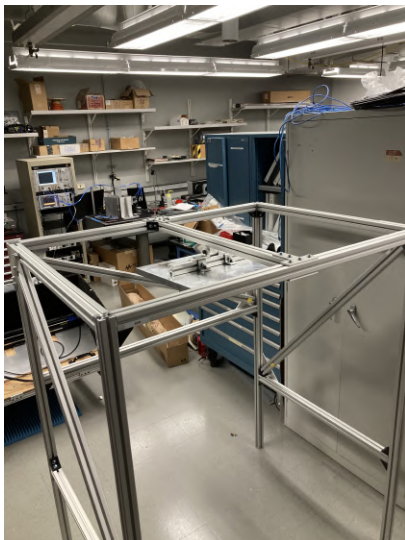
(a) One side of the fold mirror structure



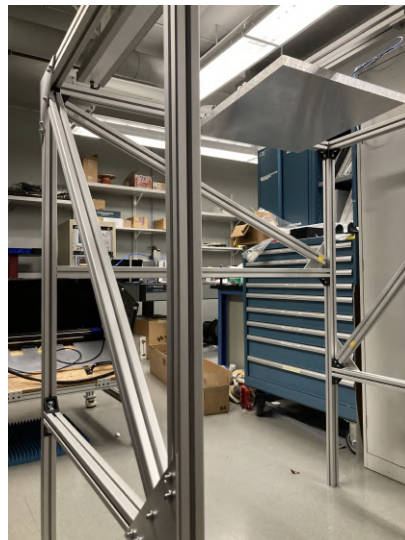
(b) Global frame assembly of the fold mirror structure

Figure 5.4: First steps of building the fold mirror structure

After adding the fold mirror we got the following assembly, as shown on figure 5.5:



(a) Up view of the structure with the fold mirror attached



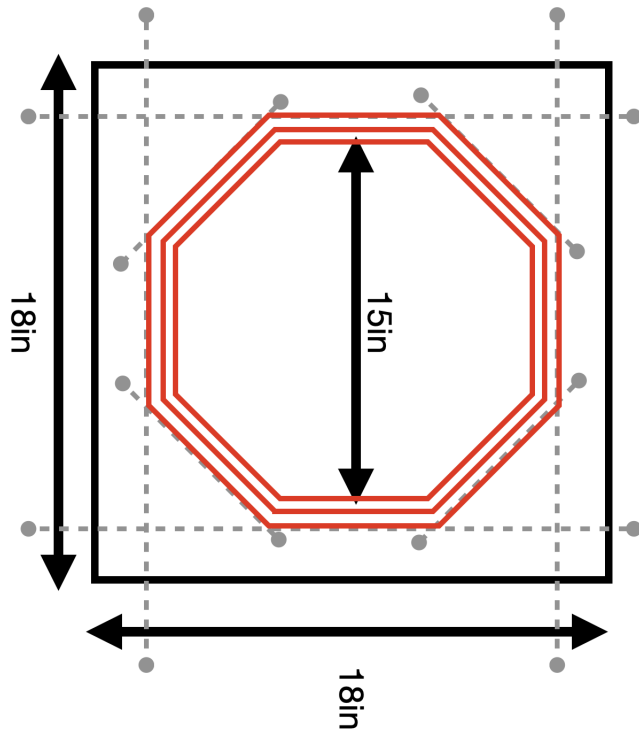
(b) Down view of the structure with the fold mirror attached



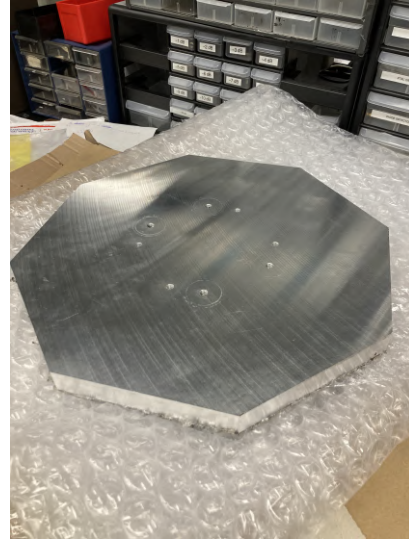
(c) Close view of the structure with the fold mirror attached

Figure 5.5: Views of the fold mirror structure from different angles

Here came the problem of the fold mirror: its weight created huge constraints on the pivot of the interface with the frame holding structure. We decided to reduce the dimensions of the MIC6 plate by cutting an octagon on the mirror according to the pattern shown on the following figure 5.6:



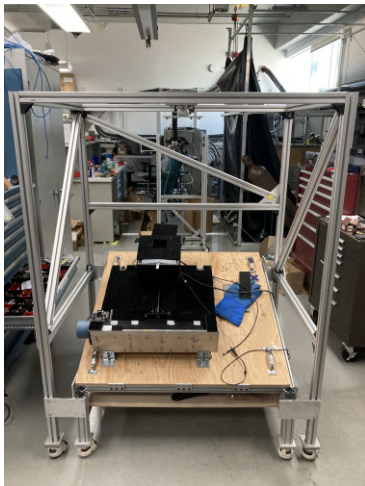
(a) Octagon pattern cut Fold Mirror



(b) Fold Mirror after cut

Figure 5.6: Octagon cut realized on the fold mirror

The cut was performed by the mechanical shop of the GALCIT - Graduate Aerospace Laboratories at Caltech and made it possible to reduce by around 46% the weight of the fold mirror. We obtain the following structure when attached to the beam mapper, shown on figure 5.7:



(a) Front view of the fold mirror / beam-mapper assembly



(b) Side view of the fold mirror / beam-mapper assembly



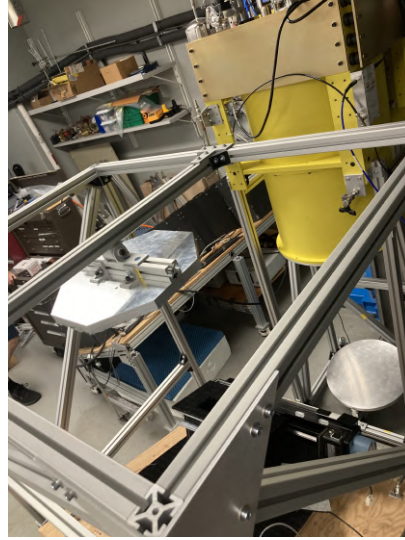
(c) Back view of the fold mirror / beam-mapper assembly

Figure 5.7: Views of the fold mirror / beam-mapper assembly from different angles

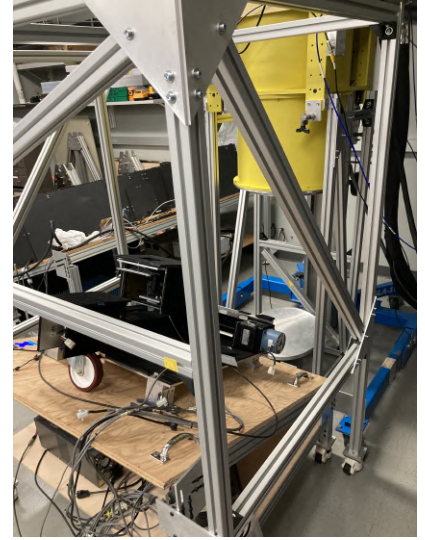
Finally, after attaching the cut fold-mirror and coupling the ellipsoidal structure to the cryostat and the fold mirror structure to the beam-mapper we can obtain the full setup which will be used in the next part for the measurement. The full setup is shown on figure 5.8:



(a) Left view of the whole optical setup



(b) Up view of the whole optical setup



(c) Right view of the whole optical setup

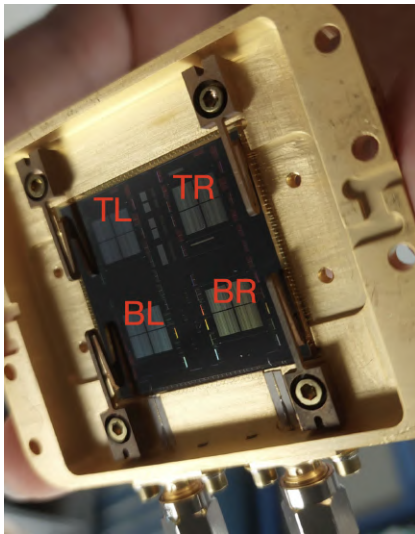
Figure 5.8: Views of the whole final optical setup from different angles

5.2 Optical Measurements

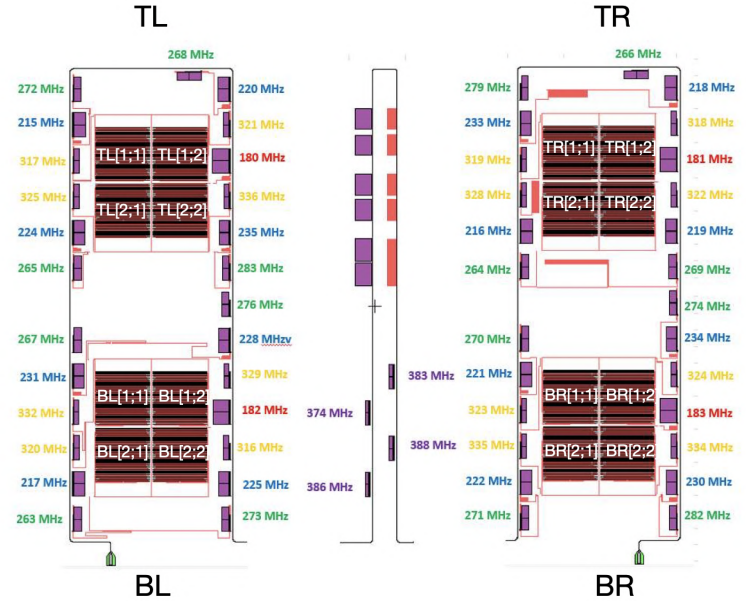
5.2.1 Study of the position of the antenna/sub-antennas along the optical path

Now the final optical setup is complete we have to perform the measurement to make sure that the image of the detectors is rebuilt and focused on the plane of the beam-mapper. For that we will first give a specific denomination to the different antennas and sub-antennas, as shown on figure 5.9 below. TL (Top-Left), TR (Top-Right), BL (Bottom-Left) and BR (Bottom-Right) are for the antennas and we will consider their sub-antennas as matrix arrays in the orientation of the device settlement in the cryostat:

$$\begin{bmatrix} \begin{bmatrix} TL[1,1] & TL[1,2] \\ TL[2,1] & TL[2,2] \end{bmatrix} & \begin{bmatrix} TR[1,1] & TR[1,2] \\ TR[2,1] & TR[2,2] \end{bmatrix} \\ \begin{bmatrix} BL[1,1] & BL[1,2] \\ BL[2,1] & BL[2,2] \end{bmatrix} & \begin{bmatrix} BR[1,1] & BR[1,2] \\ BR[2,1] & BR[2,2] \end{bmatrix} \end{bmatrix}$$



(a) Picture of the device with the denominated antennas



(b) Schematics of the device with denominated sub-antennas

Figure 5.9: Detailed view of the device with antennas and sub-antennas

To understand the expected position of each antenna and sub-antenna we have to take into account the double reflection followed by the beam through the ellipsoidal mirror and fold mirror which will invert both axis each time. As we can see on the following figure 5.10 the double axis inversion is only valid because the beam-mapper focal plane is placed **before** the second foci F2 of the ellipsoidal mirror:

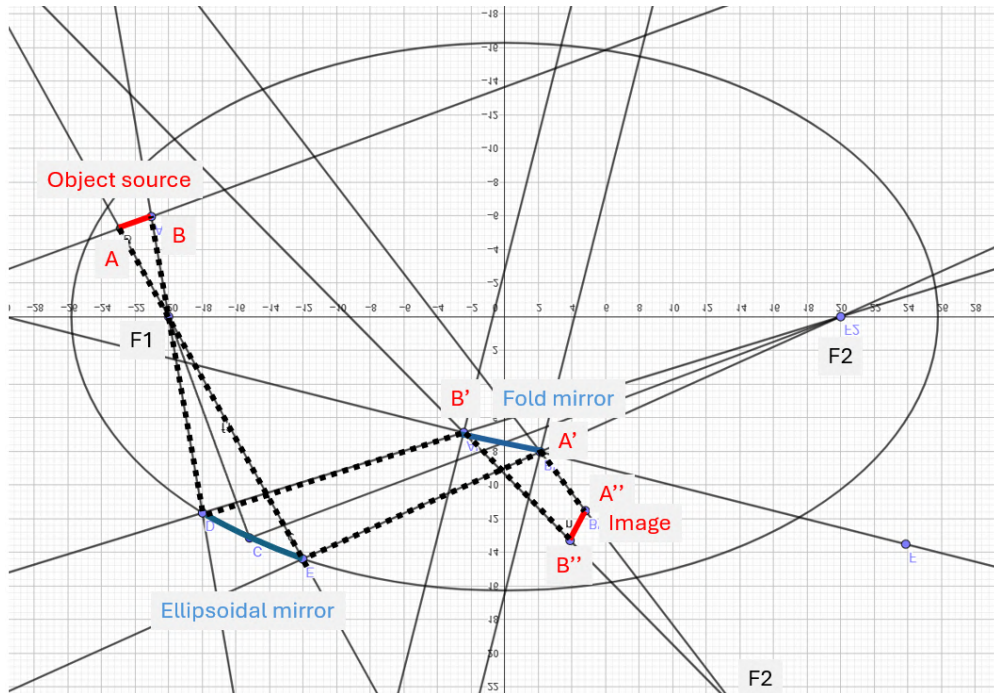


Figure 5.10: Expected image orientation after reflection on the two mirrors

We can then deduce the expected position of each sub-antenna on the beam-mapper's plane after reflections on the two mirrors as shown on figure 5.11:

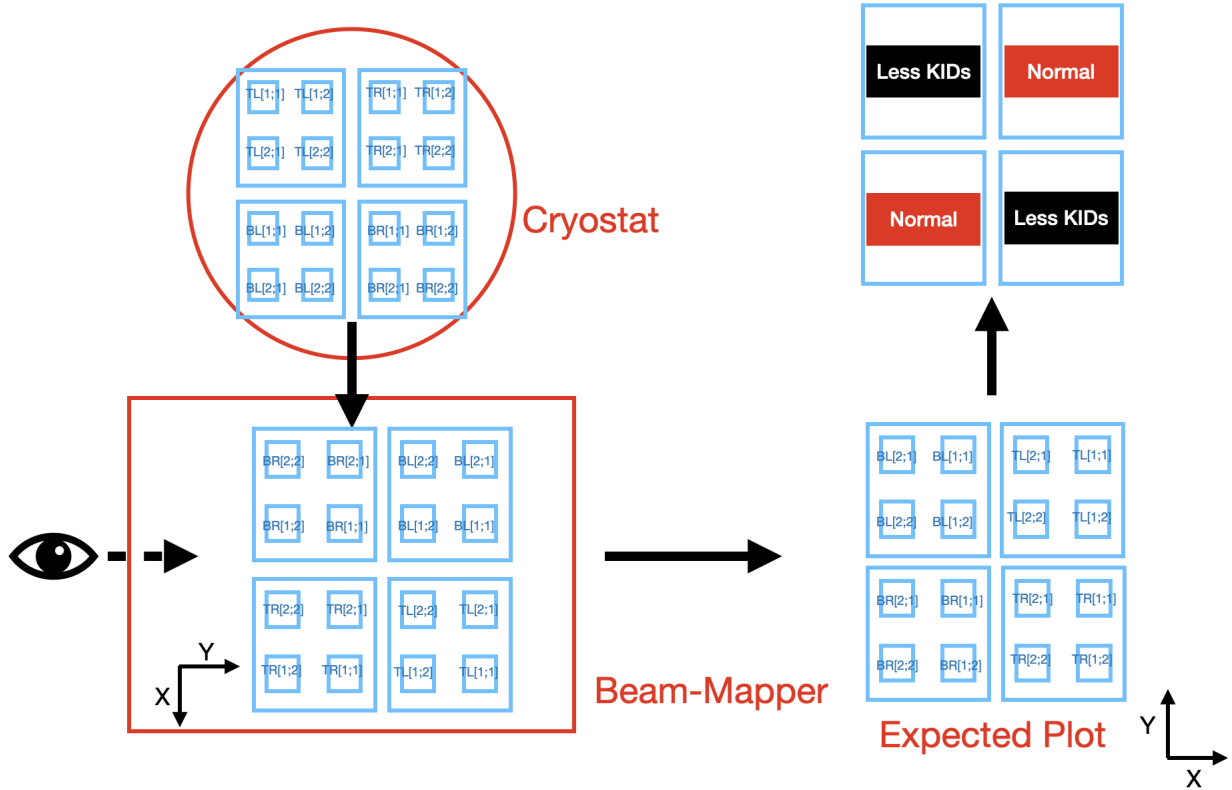


Figure 5.11: Expected positions of each sub-antennas at the end of the optical path

Each antenna is characterized as following:

- TL: Normal antenna
- TR: Modified circuit (Longer microstrip), this antenna should provide less data on the measurement plots as less KIDs are connected to this antenna
- BL: Modified circuit (Fabry Perrot cavity), this antenna should also provide less data on the measurement plots as less KIDs are connected to this antenna
- BR: Normal antenna

According to the deductions made earlier we should expect the following plot:

$$\begin{bmatrix} LOW & NORMAL \\ NORMAL & LOW \end{bmatrix}$$

5.2.2 Results of the experiments

In this part we will have a look to the results taken from the detectors and compare the position of each pixel. For the experiment we launch the measurements 5 times, each time with a different set of frequency which correspond to the frequencies of the KIDs. We call

these batch the tones numerated from 1 to 5. Each measurement with these frequencies makes it possible to get the position of some of the components along the beam-mapping plane. By overlaying all the pixels we can rebuild the image of the detector, as shown in the following figure 5.12:

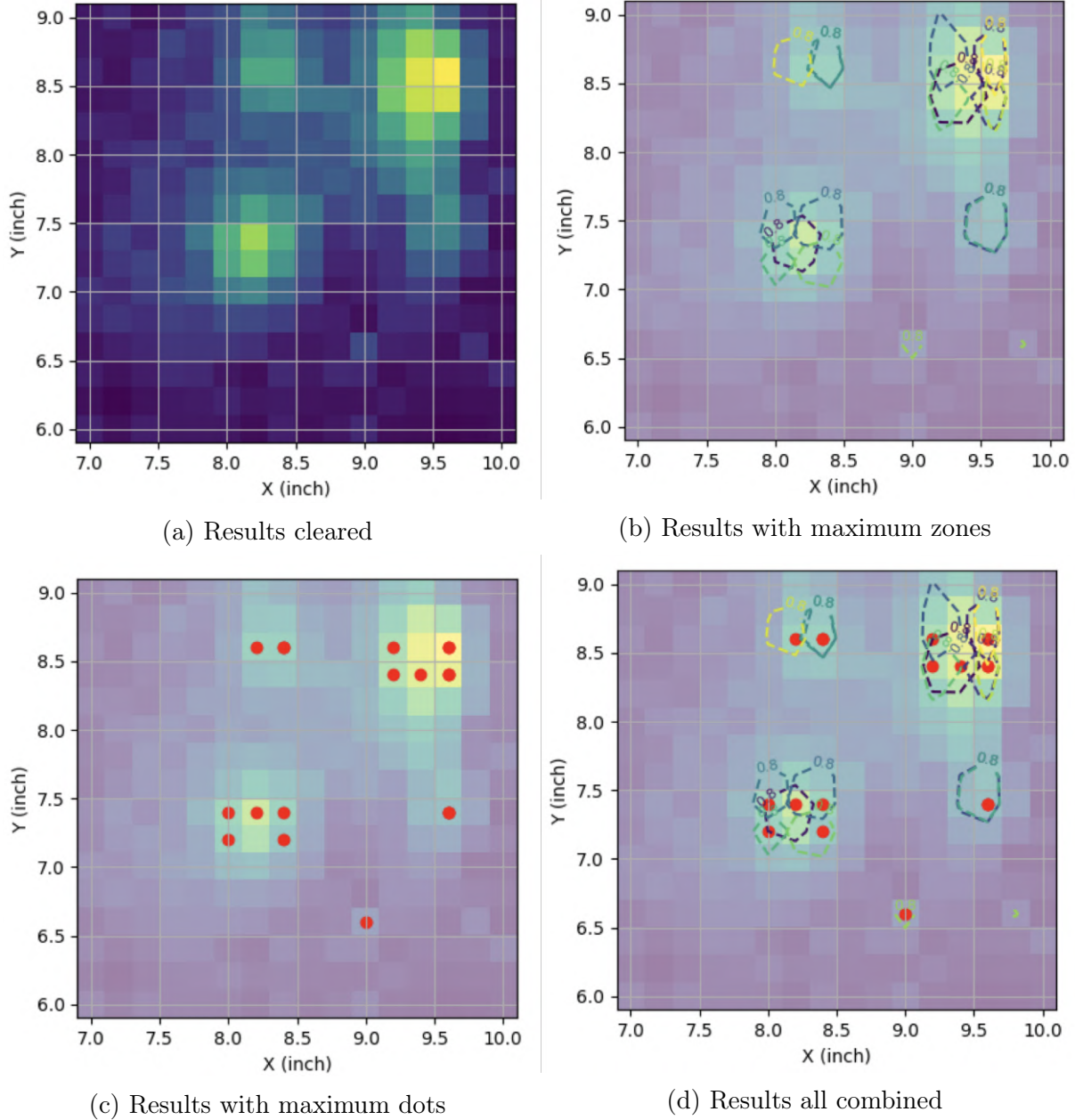


Figure 5.12: Results of the beam-mapper measurements

As we can see on figure 5.12, two antennas are clearly visible on the top right and bottom left. These pixels correspond to the antennas with a normal circuit (see previous subsection). The two pixels on the top left and bottom right are less visible and correspond to the antennas with the modified circuits. We can then recognize each antennas as their image are rebuilt on the beam-mapper plane. More importantly, the calculated zones of maximum on the two normal antennas shows relatively clearly the presence of four

maximums distributed in a square shape. These corresponds to the sub-antennas. This means that not only can we recognize the position of the antennas but also the sub-antennas, which was the main objective of this project. Also these measures seem to validate the expectations formulated earlier.

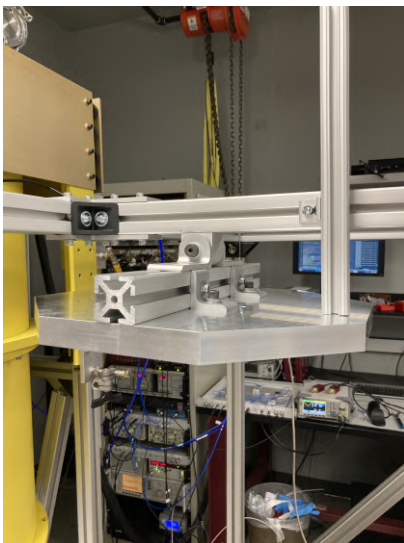
5.2.3 Repeatability of the measures

At this point of the project the main encountered problem was the repeatability of the measures: if the positions of the antennas and sub-antennas relative to each other were made possible thanks to these measures, these positions on the beam-mapping plane could slightly change due to incertitude in the removal/resettlement of the structure.

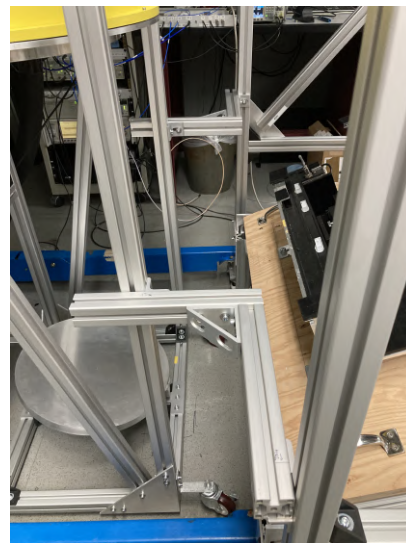
I identified some factors which could lead to fluctuations in the positions of the pixels:

- The angle of the fold mirror is slightly changed due to its weight
- the position of the assembly fold mirror and beam-mapper relatively to the assembly of the ellipsoidal mirror and cryostat is slightly modified when the assembly is removed and put back The height of the cryostat relatively to the ground can also be different because of the tolerance in the lifting machine

To cope with these possibilities I made a guidance bar on the lifting machine to ensure that the cryostat remains at the same exact height. Also I made a guidance frame on the fold mirror and designed a more rigid framed L-corner to attach the two assemblies:



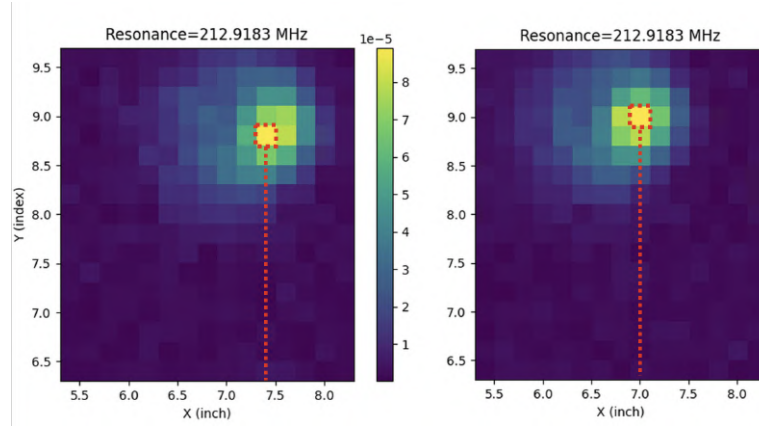
(a) Guidance frame on the fold mirror to ensure a constant angle



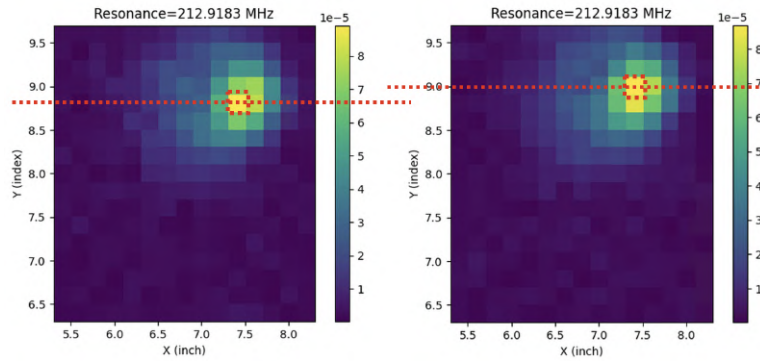
(b) L-Corner guidance frame between the two assemblies

Figure 5.13: Modifications on the structure to ensure fixed distances and orientations

With the modifications shown on figure 5.13 we can perform the measurements again. For that we attached first the assemblies, performed a first campaign of measures with one frequency, removed the assemblies, put them back and performed the measures again with the same frequency. By doing that we can assess if the measures have a good repeatability and if we get the same positions. We got the following results shown on figure 5.14:



(a) Deviation on X between the two measures



(b) Deviation on X between the two measures

Figure 5.14: Deviations on X and Y between the two measures

We get here a deviation of 0.4in on X and 0.1in on Y. If the stability is not perfect and could be more precised, it still gets improved by the modifications performed on the structure. As a result the structure can rebuild the image from each antennas and sub-antennas making it possible to characterize the detector and have a satisfactory repeatability in the measures.

Conclusion

Despite some imperfections in the repeatability of the measurements, the possible identification of the antennas and sub-antennas clearly shows that the structure has successfully achieved the primary objective of this project: reconstructing the images of the detector components outside of the cryostat. This project not only represents a technical accomplishment but also provides an important step toward the precise characterization of future generations of detectors for astronomy at Caltech.

Throughout the course of the project, I was consistently struck by how fascinating and intricate the challenge turned out to be. One particularly interesting aspect was the unexpected complexity of aligning the optical structures with precision and ensure that they are easily reproducible. While I had anticipated the physical requirements of constructing the structure, I had not fully appreciated how even slight deviations in the positioning could affect the overall results. This gave me new insight into the level of detail and fine-tuning required to optimize optical reconstruction for future astronomical applications.

Additionally, the iterative process of simulating, measuring, refining, and adjusting the setup revealed nuances in the behavior of the system that were initially unforeseen. These discoveries not only deepened my understanding of the technical constraints but also made the project far more engaging than I originally anticipated. Each challenge encountered opened up opportunities for learning, eventually making the project as much an exploration of unknowns as a solution to a defined problem.

This work paves the way for future improvements in detector technology and sets a solid foundation for more advanced astronomical observations. I am excited to see how these developments will contribute to our understanding of the universe.

I would like to extend my sincerest thanks to the entire teams at Caltech and JPL for their dedication and invaluable contributions to my work. This project would not have been possible without their expertise, collaboration, and support at every step. Thank you for making my journey both enriching and rewarding.

Appendices

Appendix A: Resonant Circuits and Frequency Multiplexing

KIDs are often arranged in arrays, with each detector resonating at a slightly different frequency. This allows the signals from multiple detectors to be read out simultaneously along one feedline using frequency-domain multiplexing. A single transmission line is coupled to all the resonators, and a cryogenic amplifier can read the signals from the entire array. The following figure 5.15 shows all the resonance frequencies of all the KIDs along the feedline grouped in 5 bandwidths ranging from B1 to B5:

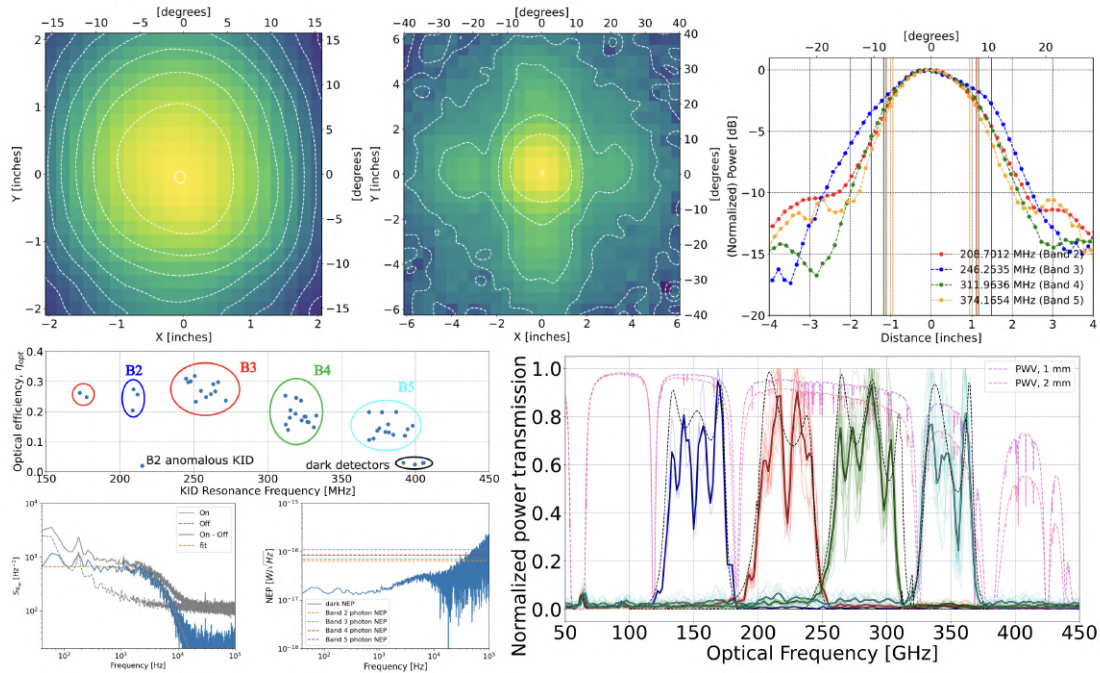


Figure 5.15: Illustration of multiplexing techniques with KIDs

On this single transmission line called the feedline all the KIDs connected have different values of geometrical inductance (depending on the dimensions of its components). This implies they have also different resonance frequencies.

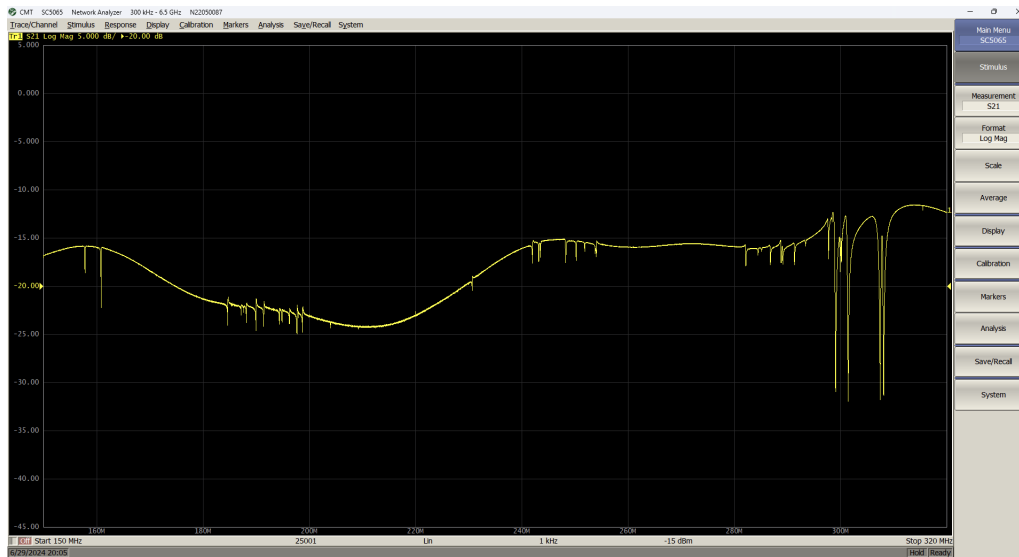


Figure 5.16: Illustration of the resonance frequencies of the KIDs along the feedline

As we can see on the plot displayed on the figure 5.16 above, on the single feedline if we move along the frequencies on the horizontal axis we can read easily the resonances of all the KIDs one by one.

Appendix B: MKID Design

The design of Microwave Kinetic Inductance Detectors (MKIDs) revolves around some important components. One of them is the resonator structure.

The resonator structure is the heart of a MKID. Each MKID is designed as a resonant circuit with a high Q-factor, enabling it to detect minute changes in inductance with exceptional accuracy. The resonant frequency of the circuit is carefully tuned to optimize sensitivity. By selecting the appropriate materials and configuration, MKIDs are able to maintain a high degree of precision in detecting very small inductive variations. An example picture of the design of a resonator is displayed on figure 5.17 below:

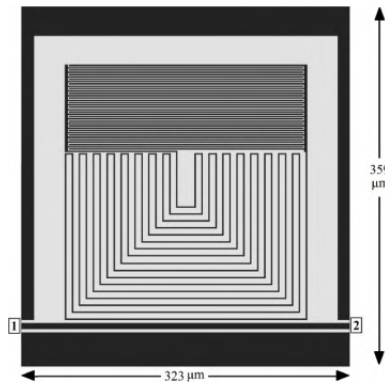


Figure 5.17: Microscopic schematics of a designed resonator [9]

As stated in the report the resonance frequency f_0 of a circuit can be described by the equation:

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

A key advantage of MKIDs is their ability to achieve a high Q-factor, where the Q-factor is given by:

$$Q = \frac{f_0}{\Delta f}$$

where Δf is the bandwidth of the resonator. Higher Q-factors allow for the detection of smaller changes in the resonant frequency, which is crucial for sensitivity in detecting faint astronomical signals.[9]

Appendix C: Advantages of MKIDs

Microwave Kinetic Inductance Detectors (MKIDs) offer several notable advantages over traditional detector technologies, particularly in astrophysical contexts. Their unique design and operation allow for high sensitivity, low noise, efficient multiplexing, and broad spectral coverage.

One of the most significant advantages of MKIDs is their high sensitivity and low noise. The use of superconducting materials enables MKIDs to detect extremely low-energy photons with minimal noise interference. This is especially important when observing faint astronomical objects, where background noise could otherwise mask critical data. Moreover, MKIDs are typically operated at cryogenic temperatures, often below 1 Kelvin, which further reduces thermal noise and enhances sensitivity. The reduction of thermal fluctuations is vital for maintaining the detector's precision, allowing it to achieve the high performance needed for advanced observations.

In terms of multiplexing capabilities, MKIDs excel in handling large-scale arrays. Thousands of MKIDs can be read out simultaneously using a single set of electronics, a feature made possible through frequency division multiplexing. By designing the resonators to operate at slightly different frequencies, each MKID can be tuned to a unique frequency. This allows multiple resonators to be read simultaneously without interference, making MKIDs highly efficient for data collection from large detector arrays. The formula for multiplexing based on frequency can be represented as:

$$f_i = f_0 + i\Delta f$$

where f_i is the frequency of the i -th resonator, f_0 is the base frequency, and Δf is the frequency spacing between resonators.

MKIDs also offer broad spectral coverage, making them versatile for different types of astronomical observations. These detectors can be designed to operate across a wide range of wavelengths, from millimeter to submillimeter, and beyond. This versatility allows MKIDs to be used in a variety of observational settings, from studying the cosmic microwave background (CMB) to detecting exoplanets and distant galaxies.

Appendix D: Applications in Astrophysics

The exceptional characteristics of MKIDs have led to their widespread adoption in several astrophysical applications. Their sensitivity, fast response times, and multiplexing capabilities make them particularly well-suited for observing the early universe and distant celestial objects.

One prominent application of MKIDs is in Cosmic Microwave Background (CMB) observations. The CMB represents the remnant radiation from the Big Bang and contains valuable information about the early structure and evolution of the universe. MKIDs are ideal for detecting the tiny fluctuations in the CMB's temperature and polarization, which can provide insights into the distribution of matter in the early universe. [10] The ability to detect these minute variations with high precision is essential for improving our understanding of cosmology and the processes that shaped the universe. A picture of the CMB can be seen on the following figure 5.18:

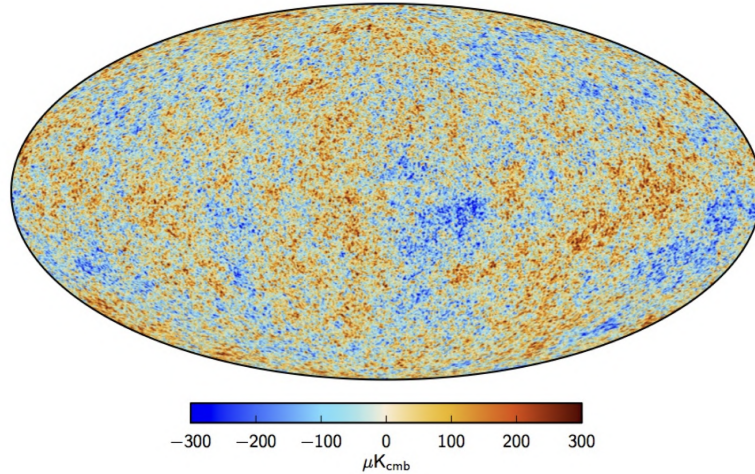


Figure 5.18: Cosmic Wave Background picture [10]

In galaxy and exoplanet surveys, MKIDs play a crucial role in detecting faint, distant objects. For deep-field galaxy surveys, MKIDs enable the detection of faint galaxies at great distances, contributing to our understanding of galaxy formation and evolution. Their ability to distinguish faint galaxies from background noise is critical for mapping the structure of the universe on a large scale.

For exoplanet imaging, MKIDs' high sensitivity and fast response times make them highly effective for direct imaging of exoplanets. Detecting exoplanets is particularly challenging due to the faintness of planets compared to the brightness of their host stars. However, MKIDs' advanced detection capabilities allow astronomers to identify and study these distant worlds, providing crucial information about their composition and potential habitability.

Appendix E: Technical Challenges and Solutions

While MKIDs offer many advantages, they also face several technical challenges that must be addressed to maximize their performance. These challenges include cryogenic cooling, frequency management, and material optimization.

Cryogenic cooling is essential for maintaining the superconductivity of the materials used in MKIDs. These detectors require extremely low temperatures, often achieved using dilution refrigerators. While these systems are highly effective, they are also complex and costly. However, ongoing developments in cryostat design are making them more efficient and reliable. Researchers are continually working on improving the thermal stability and cost-effectiveness of these systems to enhance the overall performance of MKID arrays.

Frequency management is another significant challenge, particularly in large MKID arrays where multiple resonators operate simultaneously. To prevent cross-talk between detectors, careful spacing of the resonator frequencies is required. Frequency division multiplexing techniques are employed to address this issue, ensuring that each resonator operates at a distinct frequency without interference from neighboring resonators.

Material optimization is also a key area of research. The choice of superconducting material and its fabrication can have a significant impact on the performance of MKIDs. Different materials offer various levels of sensitivity, noise reduction, and response times. Ongoing research in superconductor materials aims to optimize these characteristics to improve the efficiency and reliability of MKIDs. The use of advanced fabrication techniques, such as thin-film deposition and nanostructuring, allows for precise control over the properties of the materials, further enhancing the performance of the detectors.

Appendix F: Calculations Ellipsoid

Here let's break down the conversion process followed to get the two Zemax input values R and K.

1. Understanding Zemax Parameters:

- **Radius of Curvature (R):** This is the distance from the mirror's vertex to the center of curvature, where the mirror's surface would form a complete sphere if extended, as visible on the schematic of the following figure 5.19. [11]

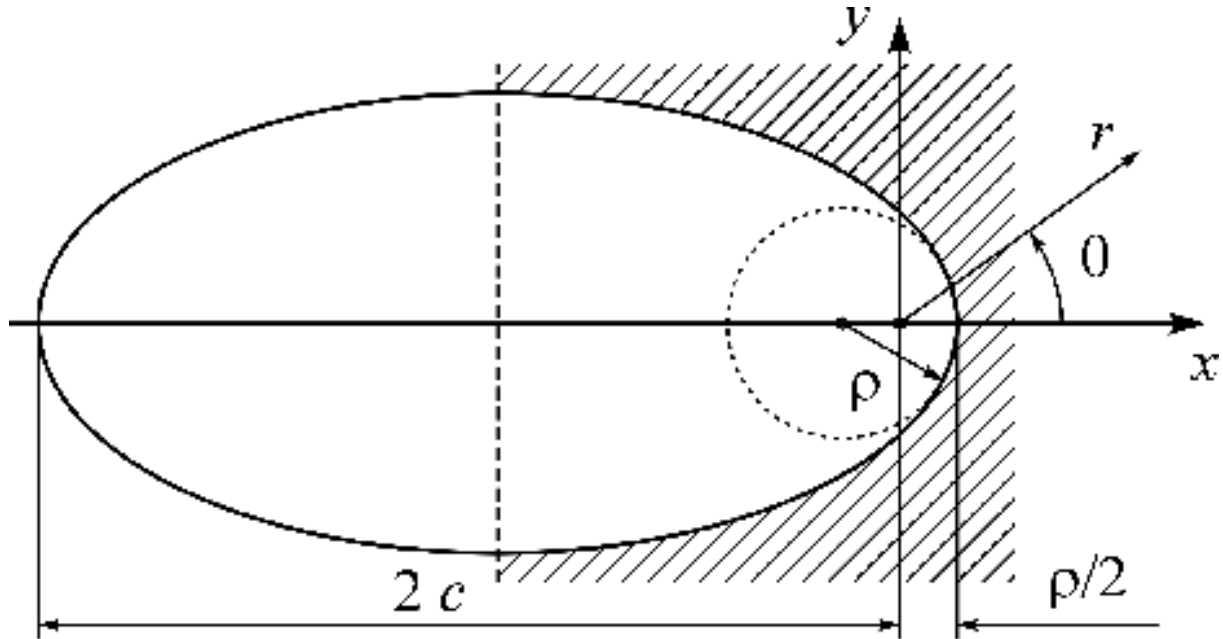


Figure 5.19: Schematics of the radius of curvature [11]

- **Conic Constant (K):** This parameter describes the mirror's deviation from a spherical shape. A conic constant of 0 corresponds to a spherical mirror, while negative and positive values indicate hyperbolic and elliptical deviations, respectively. The following figure 5.20 shows the different categories of conics and their associated constants K. We know that we will have here $-1 < K < 0$. [12]

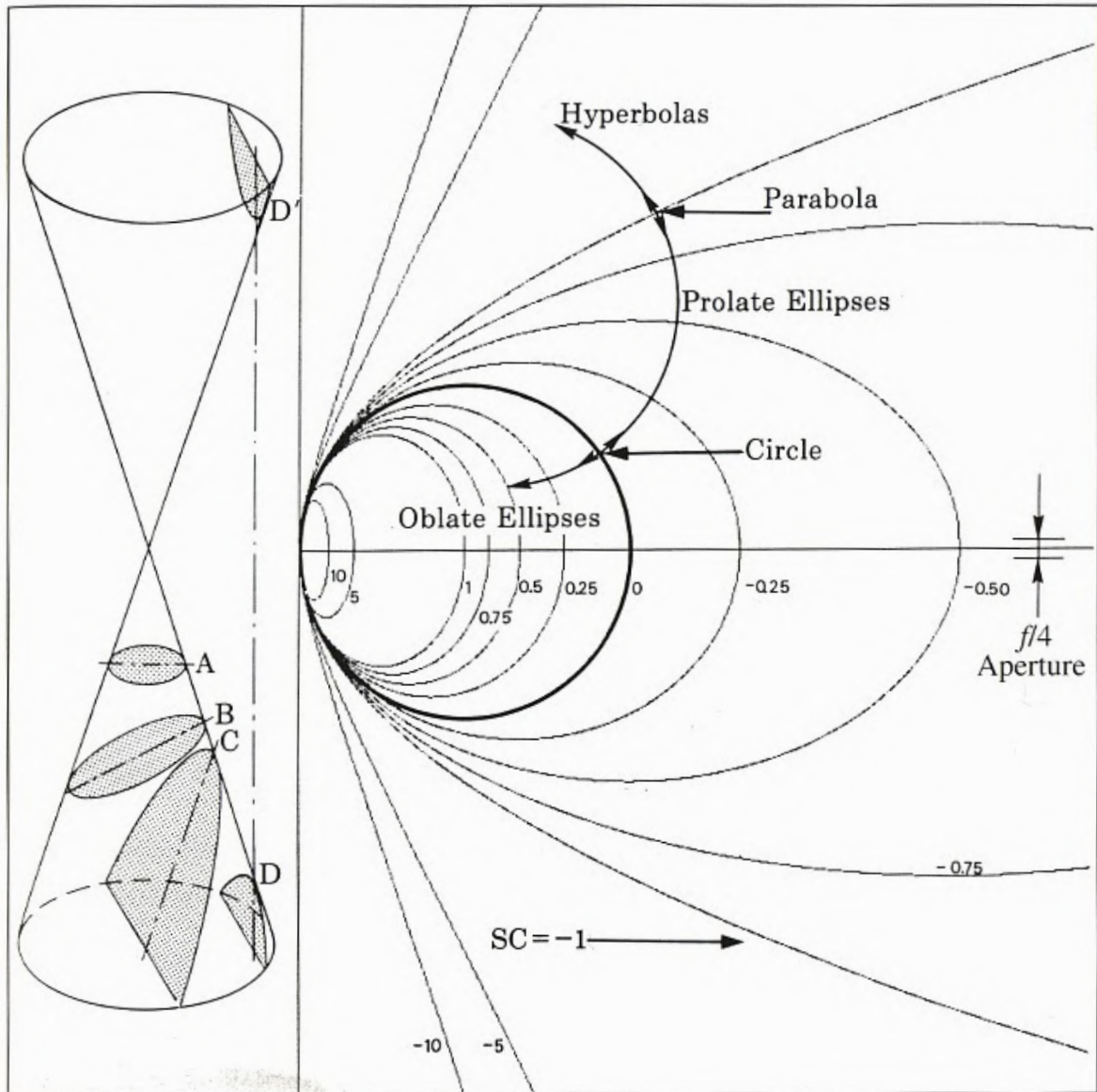


Figure 5.20: Schematics of the conics [12]

2. Conversion Process: The conversion from CAD parameters (semi-major axis a , semi-minor axis b , and eccentricity e) to Zemax parameters (R and K) involves a series of geometric and algebraic manipulations. Given the ellipsoid's equation in standard form, one can derive expressions for the radius of curvature and the conic constant in terms of a , b , and e .
3. Semi-Major and Semi-Minor Axis to Radius of Curvature (R): The radius of curvature at the vertex of an ellipsoidal mirror, considering its orientation and focal properties, can be derived from the semi-major and semi-minor axes. This derivation often involves considering the mirror's focal points and the specific segment of the ellipsoid used in the optical system. The general equation of the ellipse is given by the formula on the following figure 5.21:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

x and y are known points on the
 x and y coordinates respectively.

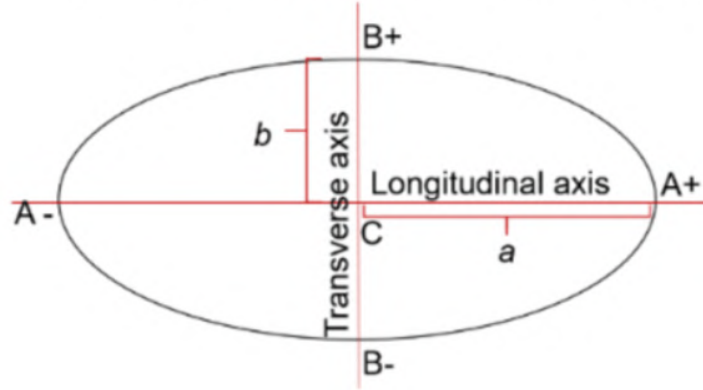


Figure 5.21: Equations of the ellipse [13]

And the formula for finding the radius of curvature at any point is given as:

$$R = \left| \frac{(1 + \left(\frac{dy}{dx}\right)^2)^{3/2}}{\frac{d^2y}{dx^2}} \right|$$

So, by using the formula in the angular system of coordinate and taking:

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

We get a formula for the radius of curvature of the ellipse as a function of the inclination angle θ :

$$R(\theta) = \frac{(b^2 \cos^2 \theta + a^2 \sin^2 \theta)^{3/2}}{ab}$$

To calculate the radius for our example we need the inclination angle of our mirror among the two foci.

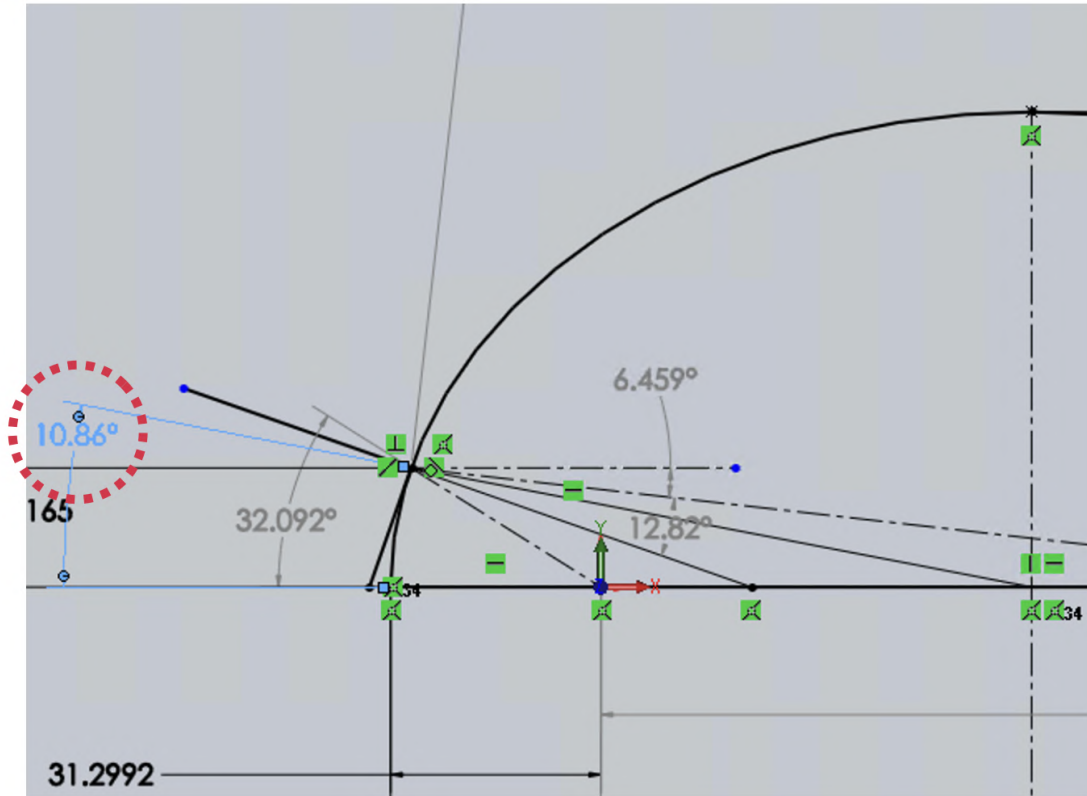


Figure 5.22: Angle of inclination of the ellipsoidal mirror

Here the angle of inclination is 10.86° hence $R = 54.643$ inches

4. Eccentricity to Conic Constant (K): The conic constant is directly related to the ellipsoid's eccentricity, a measure of how much the conic section deviates from being circular. By applying the ellipsoid's geometric definitions, one can formulate an expression that relates the eccentricity to the conic constant, facilitating the conversion necessary for Zemax modeling. Then the conic constant is given by:

$$K = \frac{-e^2}{1 - e^2} = -0.823$$

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